

Simple Threshold (Fully Homomorphic) Encryption From LWE With Polynomial Modulus

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Abstract. The learning with errors (LWE) assumption is a powerful tool for building encryption schemes with useful properties, such as plausible resistance to quantum computers, or support for homomorphic computations. Despite this, essentially the only method of achieving threshold decryption in schemes based on LWE requires a modulus that is superpolynomial in the security parameter, leading to a large overhead in ciphertext sizes and computation time.

In this work, we propose a (fully homomorphic) encryption scheme that supports a simple t -out-of- n threshold decryption protocol while allowing for a polynomial modulus. The main idea is to use the Rényi divergence (as opposed to the statistical distance as in previous works) as a measure of distribution closeness. This comes with some technical obstacles, due to the difficulty of using the Rényi divergence in decisional security notions such as standard semantic security. We overcome this by constructing a threshold scheme with a weaker notion of one-way security and then showing how to transform any one-way (fully homomorphic) threshold scheme into one guaranteeing (selective) indistinguishability-based security.

1 Introduction

In a public key encryption (PKE) scheme, one needs the secret key sk to decrypt an encrypted message. Giving one single party control of the whole secret key can be seen as a single point of failure. The study of PKE with threshold decryption aims to mitigate this by splitting the secret key into n key shares $sk_1, \dots, sk_n$, such that several key shares are needed to be able to decrypt ciphertexts. This is known as threshold public key encryption (ThPKE). In the common t -out-of- n setting, any set of t parties or fewer learns no information about encrypted messages, while any set of $t + 1$ parties can jointly decrypt ciphertexts. To decrypt, the parties first compute their own partial decryption shares and then combine them together to recover the encrypted message. When $t = n - 1$, we call it full-threshold decryption.

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Recently, NIST announced the standardization of the first cryptosystems to provide security even in the presence of quantum computers.¹ Among the finalists to be standardized, a majority base their security on the presumed hardness of (structured) lattice problems, such as Dilithium [Lyu+20] and Kyber [Sch+20] based on the (module) learning with errors problem (M-LWE) [LS15]. NIST also just began a project on threshold cryptography,² which aims to produce guidelines and recommendations for implementing threshold cryptosystems.

It is thus a very important research question to study the possibility of thresholdizing lattice-based PKE schemes. This line of research has been initiated by [BD10], where they proposed a threshold key generation and decryption starting from Regev’s encryption scheme [Reg05]. To split the secret key they use replicated secret sharing, which has a complexity that scales with $\binom{n}{t}$. Later, it has been shown that we can even build full-threshold decryption for fully homomorphic encryption (FHE) schemes [Ash+12]. A threshold fully homomorphic encryption scheme (ThFHE) allows to perform arbitrary computations on encrypted data and afterwards to partially decrypt the outcome of the computations. Their results have then been extended to t -out-of- n threshold and other access structures [Bon+18].

All works above have in common that they use a technique called *noise flooding* to guarantee that partial decryption shares do not leak any information on the underlying secret key. More precisely, each party first computes a “noiseless” partial decryption of a ciphertext using their secret key share. The noiseless partial decryptions allow recovering the message, but also reveal a small noise term e_{ct} that depends on the given ciphertext and the secret key. To prevent this leakage, every party locally adds some fresh noise on their decryption share before they jointly combine the necessary number of shares to recover the message. After decryption, the revealed noise term becomes $e_{\text{ct}} + e'$, where $e' \leftarrow \mathcal{D}_{\text{flood}}$ is a noise term that is hidden to the adversary. When proving security, the real partial decryption shares are replaced by simulated ones which do not depend on the secret key, and instead reveal noise terms of the form $e' \leftarrow \mathcal{D}_{\text{flood}}$. By arguing that the statistical distance between both ways of deriving partial decryption shares is negligible, one can argue security. While this approach has the advantage of being rather simple, it has the drawback of requiring the ratio between the flooding noise and the size of the ciphertext noise e_{ct} to be superpolynomial in the security parameter. This in turn requires the LWE problem to be secure with a superpolynomial modulus-to-noise ratio, which weakens security and requires larger LWE parameters to compensate.

Recently, multi-party reusable non-interactive secure computation (MrNISC) was constructed from LWE with a polynomial modulus [Ben+21; Shi22]. This leads to a construction of full-threshold (multi-key) FHE with a polynomial modulus. It seems plausible that their construction can also be extended to build t -out-of- n threshold FHE with polynomial modulus; however, their techniques are very complex, due to a non-black-box “round-collapsing” technique based on gar-

¹ <https://csrc.nist.gov/projects/post-quantum-cryptography>

² <https://csrc.nist.gov/Projects/threshold-cryptography>

bled circuits, so unlikely to be practical. We thus started our work asking the following research question:

Is it possible to construct a fully homomorphic encryption scheme that supports a *simple* t -out-of- n threshold decryption while allowing for a *polynomial modulus*?

Our Results. We give a positive answer to this question. On a high level, we show that the simple threshold decryption technique from previous works [BD10; Bon+18] can be significantly improved by replacing the noise flooding analysis with respect to the statistical distance by one with respect to the Rényi divergence (RD). Doing so comes with the benefit of only requiring a polynomial ratio between ciphertext noise and flooding noise, hence allowing for the desired polynomial modulus. However, it comes with several additional challenges. First, the Rényi divergence fits well in search-based security notions, such as OW-CPA security³, but does not work well with decision-based security notions, such as the standard IND-CPA security.⁴ Furthermore, it is especially difficult to apply the Rényi divergence to obtain simulation-based security, as required for typical notions of threshold decryption, since a small RD between two distributions does not imply a small statistical distance.

To overcome these challenges, we define new game-based notions of OW-CPA and adaptive/selective IND-CPA security for threshold homomorphic cryptosystems, which are compatible with Rényi divergence-based proofs, whilst also giving desirable security guarantees for applications. Then, we give general transformations from OW-CPA to IND-CPA security for ThPKE and ThFHE schemes. Whereas the first transformation only applies to standard PKE and is in the random oracle model, it comes with the advantage of guaranteeing adaptive indistinguishability as well as a form of *robustness* against up to t malicious parties, with no extra cost. The second transformation is in the standard model and also applies to the fully homomorphic setting, but does only give selective indistinguishability while not giving robustness. For the latter transformation to go through, we also need the OW-CPA ThFHE scheme to be *circuit private*; while this property is often achieved using noise flooding techniques that require a large modulus, it is also possible to use bootstrapping [DS16] or GSW-style FHE [Bou+16] to obtain circuit privacy with a polynomial modulus. Finally, we also show how to construct OW-CPA schemes based on the (module) LWE assumption with a polynomial modulus.

Put together, these techniques lead to our main result of ThFHE from (module) LWE with a polynomial modulus. More precisely, in our construction the modulus q scales as $O(\sqrt{\ell})$, where ℓ is the number of partial decryption queries made by an adversary within the security game, so q is polynomial as long as ℓ is polynomially-bounded in advance.

³ OW-CPA security for PKE roughly says that given the public key and an encryption of a random message m , it is hard to guess m .

⁴ Unless a property called *public sampleability* is fulfilled [Bai+18].

What about IND-CCA security? We could likely upgrade our construction (for PKE) to be IND-CCA secure using non-interactive zero-knowledge proofs, similarly to [Dev+21a]. However, note that (adaptive) IND-CCA security is not possible for homomorphic encryption, and IND-CPA is still useful for standard PKE; indeed, [HV22] showed that an IND-CPA secure KEM suffices to prove security of TLS-1.3. Furthermore, when running TLS with ephemeral keys and no key re-use, the adversary only ever sees a single ciphertext under any public key — this is an ideal use-case for using our ThPKE construction in a threshold post-quantum TLS setting (e.g. for hardening security of a TLS server), since we only need to choose the parameters to be secure against a single decryption query.

1.1 Overview of Techniques

Defining IND-CPA security for ThFHE (Section 3). Most of the previous IND-CPA security definitions of ThFHE required the underlying FHE scheme to be IND-CPA secure and the partial decryptions to be statistically simulatable, e.g. [Bon+18]. When replacing the statistical distance by the Rényi divergence, however, we cannot prove the statistical simulation anymore and instead have to move to a game-based notion that combines the IND-CPA game and the partial decryption queries together into one single game. Here, to support homomorphic computations, we consider a game where in each partial decryption query, first some homomorphic evaluation is performed on a set of ciphertexts, before giving decryptions of the result to the adversary. *When* and *how* the adversary gets access to the partial decryption oracle within the IND-CPA game crucially impacts the strength of the achieved security. For example, one can allow the adversary to only query partial decryptions *before* seeing the challenge ciphertext. This was done in a previous version of the ThFHE scheme proposed in [Cho+22a], which also uses a Rényi divergence based analysis. Or, one can allow the adversary to only query partial decryptions on ciphertexts that do not contain the challenge ciphertext. This is what we voted for in an earlier version of this paper [BS23a]. A more realistic setting, is to provide partial decryptions of circuit evaluations that involve the challenge ciphertext(s). Again, there are multiple choices here. If the adversary sends a list of message pairs and circuits to be encrypted, evaluated and decrypted all at one in the beginning of the game, we call it *selective* IND-CPA. If the adversary can switch between challenge encryption queries and partial decryption queries, we call it *adaptive*-IND-CPA.⁵ Of course, to prohibit trivial attacks, in both adaptive and selective flavors, the partial decryption oracle refuses to answer to queries which would directly leak which message has been encrypted when computing the challenge ciphertext. Both flavors of security notion, while lacking simulation-based security, still offer a strong guarantee

⁵ The notion introduced in [JRS17] (which is also used in an updated version of [Cho+22b]) lies in between our selective and adaptive notions of IND-CPA. In their version, the IND-CPA game is split into two sequential phases, where the adversary first sends all messages to be encrypted at once and in a second phase sends all circuits to be evaluated and then partially decrypted, again at once. The latest version of [Cho+22c] uses selective IND-CPA.

in the form of *input indistinguishability*: given partial decryptions for an evaluation $f(x_1, x_2)$, where x_1 is known to the adversary and x_2 is hidden, our security games imply that the adversary cannot distinguish whether the input x_2 was used, or some other input x'_2 such that $f(x_1, x_2) = f(x_1, x'_2)$. Similar notions have been used in secure multi-party computation [MPR06; CPP16].

To further motivate our definition, we highlight that allowing partial decryption queries that involve the challenge ciphertext is critical to achieving a meaningful notion of security. In a typical use-case, the goal of using ThFHE is to compute some function $f(x_1, \dots, x_n)$, the result of which only reveals a small amount of information compared to the inputs x_i . However, in a security game it is always the challenge ciphertext that contains the hidden information, so disallowing this in partial decryption queries does not capture the desired goals. Indeed, consider the following ThFHE scheme that is obviously insecure in this setting: firstly, modify the evaluation algorithm to output not only an encryption of $f(x_1, \dots, x_n)$, but also the encryption of x_1 ; secondly, modify the partial decryption algorithm to also output partial decryptions for x_1 . Given a set of partial decryptions for $f(x_1, \dots, x_n)$, the parties will also learn x_1 which is exactly what we want to avoid. Going back to the definition of IND-CPA security for ThFHE, as the security game of [Cho+22a] only allows for partial decryption queries before seeing the challenge ciphertext, the above obviously insecure construction could actually be shown secure using their definition.

Defining OW-CPA security for ThFHE (Section 3). As mentioned above, the Rényi divergence is hard to use in the context of decision-based security notions, such as IND-CPA. We give some intuition on why this is the case in the following. The probability preservation property of RD allows us to reason about the probability of a bad event happening in two different games. Roughly speaking, this says that if D_1, D_2 are distributions such that the Rényi divergence of D_1 from D_2 is at most δ , then for any event E , it holds that $\Pr[D_1(E)] \leq (\Pr[D_2(E)] \cdot \delta)^c$, for some constant c close to 1. If the event E occurs with negligible probability in game D_2 , then we can get by with a polynomial-sized δ to argue the same holds in D_1 . However, this is inherently hard to make use of in distinguishing games like IND-CPA, where probabilities of winning are close to $1/2$.

Instead of IND-CPA security, therefore we first aim for OW-CPA security, which is easier to prove with the Rényi divergence. When defining OW-CPA in the (fully homomorphic) threshold setting, the main changes are that the adversary also obtains t shares of the secret key and has access to a bounded number of partial decryption queries. In order to avoid trivial attacks, the partial decryption oracle refuses to answer to queries which would leak too much information on the challenge messages which the adversary tries to recover. As a measurement of *too much information* we use conditional min-entropy [Dod+08]. In other words, the oracle only answers to queries if the min-entropy of the challenge message conditioned on all the previously queried circuits and circuit evaluations is not much smaller than the original min-entropy of the challenge message.

Constructing full-threshold OW-CPA-secure ThFHE (Section 5). To simplify the presentation in the introduction, we first describe our construction in the full-threshold setting and then explain how to get t -out-of- n threshold. As a starting point, we take any encryption scheme whose decryption function is nearly linear, as is the case for most LWE-based encryption schemes (including FHE). That is, for a given ciphertext ct on a message m with respect to a key pair (sk, pk) , it holds that $\langle \text{sk}, \text{ct} \rangle = m + e_{\text{ct}}$, where e_{ct} is what we earlier called decryption noise and depends on the ciphertext and the secret key.⁶

To achieve threshold decryption, we use standard additive secret sharing to split the secret key into $\text{sk}_1, \dots, \text{sk}_n$ in a setup phase. By linearity, we could simply set the partial decryption shares as $\tilde{d}_i = \langle \text{ct}, \text{sk}_i \rangle$. However, after summing all shares together, the parties recover e_{ct} , which leaks information on sk . As in previous threshold solutions for lattice-based schemes, to compute their decryption share d_i every party now locally adds to $\tilde{d}_i$ a noise term e_i which is sampled from the noise flooding distribution $\mathcal{D}_{\text{flood}}$. When summing those partial decryption shares together, the parties learn $m + e_{\text{ct}} + \sum_{i=1}^n e_i$.

To prove the OW-CPA security of our construction, we modify the security experiment such that in a first step, the answers to the partial decryption queries no longer depend on the underlying secret key sk (reflected by e_{ct}), and in a second step the secret key shares are also independent of sk . In this case, OW-CPA security of the threshold scheme is implied by the OW-CPA security of the underlying standard encryption scheme. We simulate the partial decryption noise term $e_{\text{ct}} + \sum_{i=1}^n e_i$ by sampling some independent noise $e' \leftarrow \mathcal{D}_{\text{sim}}$. As long as the Rényi divergence between the two noise distributions is bounded by a constant, we can appeal to the probability preservation property, and the negligible probability of some PPT adversary guessing the message is preserved in both games. Note that previous works always chose $\mathcal{D}_{\text{sim}} = \mathcal{D}_{\text{flood}}$, but we later exploit in Section 6 that choosing a different $\mathcal{D}_{\text{sim}}$ can lead to better parameters.

From full-threshold to t -out-of- n threshold (Section 5). When moving to the t -out-of- n setting, a natural choice is to use Shamir secret sharing. However, this leads to the problem that reconstruction is no longer addition, and instead requires multiplying the partial decryptions with Lagrange interpolation coefficients. These coefficients may be large, which in turn blows up the noise, breaking correctness. We offer two different solutions to this issue.

First, as in [Bon+18], we can use a special type of linear secret sharing scheme with binary coefficients, so that reconstruction is always a simple sum. Efficient threshold schemes with this property exist, for any n, t . We also consider a second method based on *pseudorandom secret sharing* [CDI05], which allows the parties to generate sharings of bounded, pseudorandom values without interaction. This uses replicated secret sharing, which is more expensive, but on the other hand, allows the partial decryptions to be converted into Shamir sharings

⁶ Actually, it only reveals an encoding of m , which is easy to decode as long as parameters are set accordingly.

before reconstruction. This leads to smaller partial decryptions, slightly better parameters and gives a form of robustness via Shamir error correction.

From OW-CPA to IND-CPA security, Transform 1 (Section 4). Our first transformation (Section 4.1) can be seen as the generalization of an existing OW-CPA to IND-CPA transformation in the random oracle model [HHK17] to the threshold setting. The main idea is to use the OW-CPA-secure scheme to encrypt random messages. The vector $\mathbf{x}$ composed of those random messages then serves as input to a random oracle F , whose output hides the message m we are about to encrypt. By appending the output of a second and independent random oracle G queried on the same vector $\mathbf{x}$, we make sure that no adversary can provide incorrect decryption shares without getting caught. To this end, we define in Section 3.2 two new notions of *robustness* for (passively secure) threshold public key encryption, which might be of independent interest. The length of the vector $\mathbf{x}$ provides a trade-off between the security loss of the reduction and the compactness of ciphertexts. The resulting flavor of IND-CPA is the adaptive version. We stress that, as we explain below, we apply this transformation only to plain ThPKE, not to the fully-homomorphic case.

From OW-CPA to IND-CPA security, Transform 2 (Section 4). Whereas the reduction from above is simple and tight, it has the disadvantage of needing a random oracle to mask the message m . When we consider threshold decryption in the fully homomorphic setting, we need to make sure that we can homomorphically evaluate ciphertexts. However, the use of the random oracle makes such an evaluation impossible, as there is no efficient circuit description of random oracles. We thus propose in Section 4.2 a second transformation which now is in the standard model (but does not give robustness).

The high level idea is to encrypt a message m of δ bits, is to sample a random message x and to encrypt it using the OW-CPA-secure scheme. Then, the message bits are hidden by δ hard-core bits coming from a concatenation of δ Goldreich-Levin extractors. We use the notion of unpredictable entropy [HLR07] to give a bound on how many pseudorandom bits can be extracted from this construction. We say that a message x has unpredictability entropy k if for any PPT adversary $\mathcal{A}$ the probability of finding x given $\text{Enc}(\text{pk}, x)$ is at most 2^{-k} . We can then use existing results that show that a concatenation of δ Goldreich-Levin extractors can be used to extract $k - O(\log(1/\varepsilon))$ pseudorandom bits, where ε is the desired distinguishing advantage. Those pseudorandom bits then allow us to encrypt a message such that the ciphertexts of two given messages are computationally indistinguishable. We stress that the resulting flavor of IND-CPA is only the selective version. This is due to the use of the Goldreich-Levin extractor. We wrongly claimed adaptive security in an earlier version of this paper [BS23b].

To prove this construction IND-CPA secure, we additionally need to assume circuit privacy of the underlying OW-CPA secure FHE scheme. Intuitively, this is necessary because the IND-CPA security definition says that an adversary should not be able to distinguish between the partial decryptions of a ciphertext encrypting $f(x_1, x_2)$ and those for a ciphertext encrypting $f(x_1, x'_2)$ for some

$x'_2 \neq x_2$ where $f(x_1, x_2) = f(x_1, x'_2)$. If, for instance, $x'_2 = x_2 \oplus 1$, this is equivalent to distinguishing between ciphertexts for $f(x_1, x_2)$ and $g(x_1, x_2)$, where the function g is defined as $g(x, y) = f(x, y \oplus 1)$. This can be seen as a circuit privacy problem, thus, intuitively, it seems that some form of circuit privacy is necessary to build IND-CPA-secure FHE.

Sample Parameters and Security Analysis (Section 6). We conclude our work by discussing how to choose concrete sample parameters for our threshold PKE scheme, when instantiating it with the lattice-based scheme Kyber [Sch+20].

As an example, to obtain 1-out-of-2 threshold decryption with a single query (e.g. for ephemeral key exchange), we can use the same parameters as Kyber1024 with a modulus increased only by a factor of 5, while supporting > 100 bits of classical hardness from our reduction. In a setting with up to 2^{32} queries, we need to use a 39-bit modulus and slightly larger module rank; this increases the ciphertext size by around 5x.

Finally, we show in Section 6.2 that using the Rényi divergence noise flooding leads to almost optimal parameters by providing an attack if the adversary gets access to slightly more partial decryptions (while fixing the flooding noise). Equivalently, the attack succeeds if slightly lower flooding noise would be used while fixing the number of partial decryption queries.

1.2 Related work

Similarly to our work, [Cho+22a; Cho+22b; Cho+22c] used the Rényi divergence to obtain threshold FHE from LWE with a polynomial modulus-to-noise ratio. By arguing that the public sampleability property applies in their setting, they directly used the Rényi divergence to prove IND-CPA security. However, their work focuses on a specific construction of ThFHE based on Torus-FHE, whereas our results are phrased generically for all encryption schemes with nearly linear decryption. Lastly, they focus on linear integer secret sharing schemes, whereas we additionally propose pseudorandom secret sharing and different ways of achieving robustness.

The Rényi divergence has seen widespread use in security proofs in lattice-based cryptography, since [Bai+18]. Replacing statistical noise flooding by Rényi noise flooding has led to a significant improvement in parameters for security reductions, for instance when proving the hardness of (structured) LWE with a binary secret [Bou+20], when designing multi-key FHE [DWF22], or more recently, in the context of lattice-based threshold signatures [ASY22]. The latter work of [ASY22] is quite similar to ours, since they also apply Rényi noise flooding to threshold FHE; however, they do not directly prove security of the threshold FHE scheme, and instead analyze the resulting threshold signature scheme directly (which is based on a search problem, so amenable to a Rényi divergence analysis). They additionally show the optimality of their noise flooding by providing an attack when a smaller noise flooding ratio is used. As the attack uses that their signature scheme is deterministic, it does not directly apply to our randomized encryption scheme. Previous works [Dev+21b; Nae+20]

have already observed that OW-CPA allows to bypass the issues caused by the Rényi divergence. However, both are in the PKE setting, whereas our work focused on the FHE setting. This required some care: it is not straightforward to define a OW-CPA notion in the fully-homomorphic setting and the standard transformation used in [Dev+21b; Nae+20] to lift one-way security to indistinguishability is not suited for the fully-homomorphic setting neither.

In an independent line of work, another noise flooding technique, called gentle noise flooding, has been studied in order to avoid the superpolynomial parameter blow-up [BD20a]. It was first used in theoretical hardness results on entropic (structured) LWE [BD20a; BD20b]. Later, a similar technique was used in [Cas+22] for improving parameters in additively homomorphic encryption with circuit privacy. The setting of [Cas+22] is quite different to ours, however, since with circuit privacy, the challenge is to deal with leakage on a plaintext rather than the secret key. This is handled via gentle noise flooding by applying a randomized encoding to the plaintext, so that leaking a constant fraction of its coordinates does not reveal anything about the plaintext. A similar technique does not seem to work in the threshold setting, with leakage on the secret key.

From a high level perspective, our adaptive notion of IND-CPA security has some similarities to the notion of IND-CPA^D security introduced in [LM21] in the context of approximate FHE. For instance, partial decryption queries in our setting correspond to decryption queries in their setting. Our security notion further matches with the game-based input-indistinguishability notion in the context of secure multi-party computation from [MPR06; CPP16], when realizing the latter with the help of ThFHE.

Another approach to build threshold key generation and decryption protocols is to use general multi-party computation tools like garbled circuits. This was done in [Kra+19] for a Ring-LWE based scheme. Their solution does not need any noise flooding or increased parameters of the underlying scheme, however, it relies on generic multi-party computation techniques like garbled circuits, and the partial decryption shares are generated using an expensive, interactive protocol rather than non-interactively as in our setting.

1.3 Changelog

This paper has been updated multiple times since its first apparition on the IACR ePrint server. To better help navigating the different versions, we give a quick summary of the changes below.

- [BS23a] We correctly used **weak- ℓ -IND-CPA** security (Def. 25).
- [BS23b] After having observed, that **weak- ℓ -IND-CPA** is not a very realistic security notion, we switched to **adaptive- ℓ -IND-CPA** security (Def. 20) for both OW-CPA to IND-CPA transformations in Section 4.
- [BS23c] An issue with the proof of Theorem 3 was found by anonymous reviewer(s). As a fix, we added the circuit privacy condition (also for the published version [BS23e]).

- [BS23d] We were made aware by the authors of [Acp] that the proof of Theorem 3 is not correct as it is. We don't obtain adaptive- ℓ -IND-CPA security, but only selective- ℓ -IND-CPA security for the transformation in Section 4.2. The transformation in Section 4.1 still leads to adaptive- ℓ -IND-CPA security.
- This version: We slightly reformulated the proof of Lemma 13, making it more formal.

2 Preliminaries

For any positive integer q , we denote by $\mathbb{Z}_q$ the integers modulo q and for any positive integer n , we denote by $[n]$ the set $\{1, \dots, n\}$. Vectors are denoted in bold lowercase and matrices in bold capital letters. The identity matrix of order m is denoted by $\mathbf{I}_m$. The concatenation of two matrices $\mathbf{A}$ and $\mathbf{B}$ with the same number of rows is denoted by $[\mathbf{A}|\mathbf{B}]$. The abbreviation PPT stands for probabilistic polynomial-time. When we split a PPT adversary $\mathcal{A}$ in several sub algorithms $(\mathcal{A}_i)_{i \in [n]}$, we implicitly assume that $\mathcal{A}_i$ outputs a state that is passed to the next $\mathcal{A}_{i+1}$. We call a function $\text{negl}(\cdot)$ negligible in λ if $\text{negl}(\lambda) = \lambda^{-\omega(1)}$, i.e., it decreases faster towards 0 than the inverse of any polynomial.

Throughout the paper we make use of the random oracle model (ROM), where we assume the existence of perfectly random functions, realized by oracles. For a random oracle $F: \{0, 1\}^n \rightarrow \{0, 1\}^m$ it holds that $\Pr[F(x) = y] = 2^{-m}$ and that $\Pr[F(x) = F(x') = y: x \neq x'] = \Pr[F(x) = y] \cdot \Pr[F(x') = y] = 2^{-2m}$. Hence, random oracles are per definition collision resistant. For $x, y \in \{0, 1\}^n$ we denote by $x \oplus y$ the bit-wise XOR operator.

2.1 Probability and Entropy

For a finite set S , we denote its cardinality by $|S|$ and the uniform distribution over S by $U(S)$. The operation of sampling an element $x \in S$ according to a distribution D over S is denoted by $x \leftarrow D$, where the set S is implicit.

For standard deviation $\sigma > 0$ and mean $c \in \mathbb{R}$, we define the continuous Gaussian distribution $D_{\sigma,c}: \mathbb{R} \rightarrow (0, 1]$ by $D_{\sigma,c}(x) = 1/(\sigma\sqrt{2\pi}) \cdot \exp(-(x - c)^2/(2\sigma^2))$. We also define the rounded Gaussian distribution over $\mathbb{Z}$, by rounding the result to the nearest integer, and denote this by $\lfloor D_{\sigma,c} \rfloor$.

A random variable X over $\mathbb{R}$ is called τ -subgaussian for some $\tau > 0$ if for all s it holds $\mathbb{E}[\exp(sX)] \leq \exp(\tau^2 s^2/2)$. A τ -subgaussian random variable satisfies $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^2] \leq \tau^2$. We associate to X the width $\sigma = \sqrt{\mathbb{E}[X^2]}$. The continuous Gaussian distribution D_σ and its rounded version $\lfloor D_\sigma \rfloor$ are σ -subgaussian. Further, the uniform distribution over $[-a, a] \cap \mathbb{Z}$ is a -subgaussian.

The statistical distance between two probability distributions X and Y , denoted by $\text{sdist}(X, Y)$, is defined as $\max_T |\Pr[T(X) = 1] - \Pr[T(Y) = 1]|$, where T is any test function. The computational distance with respect to size s circuits, denoted by $\text{cdist}_s(X, Y)$, limits T to be any circuit of size s . For any event E , the probability preservation property of sdist (resp. cdist_s) states that $X(E) \leq Y(E) + \text{sdist}(X, Y)$ (resp. $X(E) \leq Y(E) + \text{cdist}_s(X, Y)$).

The notion of unpredictable entropy has been introduced and studied in [HLR07] in the context of conditional computational entropy.

Definition 1 (Unpredictable Entropy). *For a distribution (X, Z) , we say that X has unpredictable entropy at least k conditioned on Z , if there exists a collection of distributions Y_Z (giving rise to a joint distribution (Y, Z)) such that $\text{cdist}_s((X, Z), (Y, Z)) \leq \varepsilon$, and for all circuits C of size s ,*

$$\Pr[C(Z) = Y] \leq 2^{-k}.$$

We write $H_{\varepsilon, s}^{\text{unp}}(X|Z) \geq k$.

Definition 2 (Concatenated Goldreich-Levin Extractor). *Fix $n, \delta \in \mathbb{N}$. We define the concatenated Goldreich-Levin extractor $\mathcal{E}: \{0, 1\}^n \times (\{0, 1\}^n)^\delta \rightarrow \{0, 1\}^\delta \times (\{0, 1\}^n)^\delta$ as*

$$\mathcal{E}(x, s_1, \dots, s_\delta) := (\langle x, s_1 \rangle \bmod 2, \dots, \langle x, s_\delta \rangle \bmod 2, s_1, \dots, s_\delta).$$

Definition 3 (Reconstruction procedure [HLR07, Def. 6]). *Let $E: \{0, 1\}^n \times \{0, 1\}^d \rightarrow \{0, 1\}^m \times \{0, 1\}^d$ be a function whose last d outputs equal the last d input bits. E has (ℓ, ε) -reconstruction if there is a pair of oracle algorithms (C, D) , where $C(\cdot): \{0, 1\}^n \rightarrow \{0, 1\}^\ell$ is compressing, while $D(\cdot): \{0, 1\}^\ell \rightarrow \{0, 1\}^n$ is a “decompressor” that runs in time polynomial in n . Furthermore, for every x and distinguisher T , if $|\Pr[T(E(x, U_d)) = 1] - \Pr[T(U_m \times U_d) = 1]| > \varepsilon$ then $\Pr[D^T(C^T(x)) = x] > 1/2$.*

Lemma 1. *The concatenated Goldreich-Levin extractor $\mathcal{E}$ has $(\delta + \ell, \delta\varepsilon)$ -reconstruction for any $\varepsilon \in (0, 1]$ and $\ell = \log_2 2n + 2\log_2(1/\varepsilon)$.*

Proof. This is a consequence of the proof of the Goldreich-Levin theorem [GL89]. First, let T be a distinguisher for the Goldreich-Levin extractor with $\delta = 1$. Suppose that the advantage of T is larger than ε , that is, for any x , it holds that $|\Pr[T(\mathcal{E}(x, U_n)) = 1] - \Pr[T(U_1 \times U_n) = 1]| > \varepsilon$. From the GL theorem (see, for instance, the presentation by Bellare [Bel99, Thm. 3]), there exists an algorithm A that, given oracle access to T , runs in time $O(n^3\varepsilon^{-4})$ and outputs a list $\mathcal{L}$ of $M = 2n/\varepsilon^2$ strings, such that $\Pr[x \in \mathcal{L}] > 1/2$. Define the algorithm C to simply run A and output the index of x in $\mathcal{L}$, which is $\log_2 M$ bits, and define D to iteratively compute $\mathcal{L}$ and output the corresponding element. This shows that the GL extractor with $\delta = 1$ has (ℓ, ε) -reconstruction for $\ell = \log_2(M) = \log_2(2n/\varepsilon^2)$. For $\delta > 1$, from [HLR07, Prop. 1] we obtain that the concatenation of δ extractors with (ℓ, ε) -reconstruction has $(\delta + \ell, \delta\varepsilon)$ -reconstruction. $\square$

Using the reconstruction property of Goldreich-Levin, we get the following bound on the number of pseudorandom bits that can be extracted.

Lemma 2 ([HLR07, Lemma 6]). *Let X be a distribution with unpredictable entropy $H_{\varepsilon, s}^{\text{unp}}(X|Z) \geq k$ and let $\mathcal{E}$ be the concatenated Goldreich-Levin extractor*

for some $n, \delta \in \mathbb{N}$. If $k = \delta + \log_2 2n + 3 \log_2 1/\varepsilon$, then $\mathcal{E}$ extracts δ pseudorandom bits, i.e.,

$$\text{cdist}_{s'}((Z, \mathcal{E}(X, U(\{0, 1\}^{n\delta}))), (Z, U(\{0, 1\}^\delta \times \{0, 1\}^{n\delta}))) \leq 5\delta\varepsilon,$$

where $s' = O(sn^{-3}\varepsilon^4)$.

Let $\mathbf{x}$ follow a distribution on a set X , and $\mathbf{z}$ follow a possibly correlated distribution on a set Z . The *average conditional min-entropy* [Dod+08] of $\mathbf{x}$ given $\mathbf{z}$ is defined by

$$\tilde{H}_\infty(\mathbf{x}|\mathbf{z}) = -\log_2 \left(E_{\mathbf{z}'} \left[\max_{\mathbf{x}' \in X} \Pr[\mathbf{x} = \mathbf{x}' | \mathbf{z} = \mathbf{z}'] \right] \right).$$

Lemma 3 ([Dod+08, Lem. 2.2]). *Let $\mathbf{x}, \mathbf{y}, \mathbf{z}$ be three random variables, where $\mathbf{z}$ takes at most 2^λ values. Then*

$$\tilde{H}_\infty(\mathbf{x}|\mathbf{y}, \mathbf{z}) \geq \tilde{H}_\infty(\mathbf{x}|\mathbf{y}) - \lambda.$$

The Rényi divergence (RD) defines an alternative measure of distribution closeness. We follow [Bai+18] and use a definition of the RD which is the exponential of the classical definition. We restrict the order a to be in $(1, \infty)$.

Definition 4 (Rényi Divergence). *Let P and Q be two discrete probability distributions such that $\text{Supp}(P) \subseteq \text{Supp}(Q)$. For $a \in (1, \infty)$ the Rényi divergence of order a is defined by*

$$\text{RD}_a(P, Q) = \left(\sum_{x \in \text{Supp}(P)} \frac{P(x)^a}{Q(x)^{a-1}} \right)^{\frac{1}{a-1}}.$$

The definitions are extended in the natural way to continuous distributions. We recall some useful properties of the RD. The first two were proven in [EH14] and the last one was proven in [Ros20, Prop. 2].

Lemma 4. *Let P, Q be two discrete probability distributions with $\text{Supp}(P) \subseteq \text{Supp}(Q)$. For $a \in (1, \infty)$, it yields:*

Data Processing Inequality: $\text{RD}_a(g(P) \| g(Q)) \leq \text{RD}_a(P \| Q)$ for any function g , where $g(P)$ (resp. $g(Q)$) denotes the distribution of $g(y)$ induced by sampling $y \leftarrow P$ (resp. $y \leftarrow Q$).

Probability Preservation: Let $E \subset \text{Supp}(Q)$ be an event, then for $a \in (1, \infty)$

$$Q(E) \cdot \text{RD}_a(P \| Q) \geq P(E)^{\frac{a}{a-1}}.$$

Multiplicativity: Let P, Q be two probability distributions of a pair of random variables (Y_1, Y_2) . For $i \in \{1, 2\}$, let P_i (resp. Q_i) denote the marginal distribution of Y_i under P (resp. Q), and let $P_{2|1}(\cdot|y_1)$ (resp. $Q_{2|1}(\cdot|y_1)$) denote the conditional distribution of Y_2 given that $Y_1 = y_1$. Then for $a \in (1, \infty)$

$$\text{RD}_a(P \| Q) \leq \text{RD}_a(P_1 \| Q_1) \cdot \max_{y_1 \in Y_1} \text{RD}_a(P_{2|1}(\cdot|y_1) \| Q_{2|1}(\cdot|y_1)).$$

The Rényi divergence of two shifted Gaussians is given below. This also allows us to bound the RD of rounded Gaussians by the data processing inequality.

Lemma 5 ([GAL13]). *Let σ be a positive real number and $c \in \mathbb{Z}$. Then for $a \in (1, \infty)$ it yields*

$$\text{RD}_a(D_{\sigma,c} \| D_\sigma) = \exp\left(\frac{ac^2}{2\sigma^2}\right).$$

Lemma 6. *Let D_1, D_2 be two probability distributions over $\mathbb{Z}$ and $e_1, \dots, e_N$ be (possibly dependent) random variables over $\mathbb{Z} \cap [-B, B]$ for some $B \in \mathbb{Z}$, for which there exist $a \in (1, \infty)$ and $\rho \geq 1$ such that for all β with $|\beta| \leq B$, it holds that $\text{Supp}(D_1 + \beta) \subseteq \text{Supp}(D_2)$, and furthermore, $\text{RD}_a(D_1 + \beta \| D_2) \leq \rho$. Then,*

$$\text{RD}_a((D_1 + e_N, \dots, D_1 + e_1) \| D_2^N) \leq \rho^N.$$

Proof. We apply N times the multiplicativity property of the Rényi divergence as follows. Let $P = (D_1 + e_N, \dots, D_1 + e_1)$ and $Q = D_2^N$. Our goal is to bound $\text{RD}_a(P \| Q)$. We start with setting their marginal distributions as $P_1 = (D_1 + e_{N-1}, \dots, D_1 + e_1)$, $Q_1 = D_2^{N-1}$, $P_2 = D_1 + e_N$ and $Q_2 = D_2$. For $j \in [N]$, let E_j denote the random variable given by the distribution $D_1 + e_j$. By Lemma 4, it yields

$$\begin{aligned} \text{RD}_a(P \| Q) &\leq \text{RD}_a(P_1 \| Q_1) \cdot \max_{y_1 \in Y_1} \text{RD}_a(D_1 + e_N | Y_1 = y_1 \| D_2 | Y_1 = y_1) \\ &\leq \text{RD}_a(P_1 \| Q_1) \cdot \max_{y_1 \in Y_1} \text{RD}_a(D_1 + \beta | Y_1 = y_1 \| D_2 | Y_1 = y_1) \\ &\leq \text{RD}_a(P_1 \| Q_1) \cdot \text{RD}_a(D_1 + \beta \| D_2) \\ &\leq \rho \cdot \text{RD}_a(P_1 \| Q_1), \end{aligned}$$

where β is such that $|\beta| \leq B$ and $Y_1 = (E_{N-1}, \dots, E_1)$. From line 2 to line 3 we used the fact that neither $D_1 + \beta$ nor D_2 depend on Y_1 anymore. Finally, we obtain $\text{RD}_a(P \| Q) \leq \rho^N$ by induction. $\square$

2.2 Linear Secret Sharing

We use linear secret sharing schemes (LSSS) for monotone access structures with a special $\{0, 1\}$ -reconstruction property, as follows.

Definition 5 (Monotone Access Structure). *Let $\mathcal{P} = \{P_1, \dots, P_n\}$ be a set of parties and $2^{\mathcal{P}}$ its power set. A monotone access structure is a collection of sets $\mathbb{A} \subset 2^{\mathcal{P}}$, such that for any $S \in \mathbb{A}$, if $T \supset S$ then $T \in \mathbb{A}$. We say that $\mathbb{A}$ is efficient if membership of $\mathbb{A}$ can be verified in time $\text{poly}(\lambda)$, where $\mathbb{A}$ is viewed as a function of λ .*

In this work, we only consider efficient access structures. To ease notation, we identify a party P_i with its index i , viewing each set $S \in \mathbb{A}$ as a subset of $[n]$. For any $S \subset [n]$ and vector $\mathbf{v} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$, we let $\mathbf{v}|_S$ denote the vector of shares restricted to $\mathbf{v}_i$ for indices $i \in S$.

Definition 6 (Linear Secret Sharing Scheme). Let q, L, n be positive integers and $\mathbb{A}$ a monotone access structure. A linear secret sharing scheme LSSS for $\mathbb{A}$ is defined by a randomized algorithm $\text{Share} : \mathbb{Z}_q \rightarrow (\mathbb{Z}_q^L)^n$ and a family of deterministic algorithms $\text{Rec}_S : (\mathbb{Z}_q^L)^{|S|} \rightarrow \mathbb{Z}_q$, for $S \subseteq [n]$, which satisfy:

Privacy: For any set $S \notin \mathbb{A}$, any $x, x' \in \mathbb{Z}_q$ and $\mathbf{v} \in \mathbb{Z}_q^{L|S|}$, it holds that $\Pr[\text{Share}(x)|_S = \mathbf{v}] = \Pr[\text{Share}(x')|_S = \mathbf{v}]$.

Reconstruction: For any set $S \in \mathbb{A}$, any $x \in \mathbb{Z}_q$ and $\mathbf{v} = \text{Share}(x)$, the reconstruction algorithm outputs $\text{Rec}_S(\mathbf{v}|_S) = x$.

Linearity: For any $\alpha, \beta \in \mathbb{Z}_q$, any set S with $|S| > t$ and any share vectors $\mathbf{u}, \mathbf{v}$, it holds that $\text{Rec}_S(\alpha\mathbf{u}|_S + \beta\mathbf{v}|_S) = \alpha\text{Rec}_S(\mathbf{u}|_S) + \beta\text{Rec}_S(\mathbf{v}|_S)$.

When the set of shares is $S = [n]$, we write Rec instead of $\text{Rec}_{[n]}$.

We need the following notion of valid and invalid share sets [Bon+18].

Definition 7. Let $x \in \mathbb{Z}_q$, $(\mathbf{v}_1, \dots, \mathbf{v}_n) = \text{Share}(x)$, and write $\mathbf{v}_i = (\mathbf{v}_{i,1}, \dots, \mathbf{v}_{i,L})$. A set of pairs of indices $T \subseteq [n] \times [L]$ is an invalid set of share elements if the corresponding shares $(\mathbf{v}_{i,j})_{(i,j) \in T}$ reveal no information about x . Otherwise, we say that T is a valid set of share elements. We additionally say:

- $T \subseteq [n] \times [L]$ is a maximal invalid set of share elements if it is invalid, but for any $(i, j) \in [n] \times [L] \setminus T$, the set $T \cup \{(i, j)\}$ is a valid set of share elements.
- $T \subseteq [n] \times [L]$ is a minimal valid set of share elements if it is valid, but for any $T' \subsetneq T$, the set T' is an invalid set of share elements.

Note that in any LSSS, a valid set as defined above always allows reconstruction of the secret x . This is because an LSSS can equivalently be defined by a matrix M , such that each share element $\mathbf{v}_{i,j}$ is computed as the inner product of some row of M and $(x, r_1, \dots, r_{n-1})$, where r is the randomness used in Share . Reconstruction is possible for a given set of share elements iff the corresponding set of rows of M span the target vector $(1, 0, \dots, 0)$. This definition implies that any set of rows is either invalid — and reveals nothing about x — or valid, and allows full reconstruction. For further details, see e.g. [Bei96, Chapter 4].

Our main construction requires that the reconstruction function Rec_S takes a 0/1 combination of its inputs. In the following, we require this to hold not only for any set of shares corresponding to a valid set of parties in $\mathbb{A}$, but for any valid set of share elements. This property is equivalent to the notion of a derived $\{0, 1\}$ -LSSS, used in [JRS17].⁷

Definition 8 (Strong $\{0, 1\}$ -Reconstruction). We say that a LSSS has strong $\{0, 1\}$ -reconstruction if for any secret x and $(\mathbf{v}_1, \dots, \mathbf{v}_n) = \text{Share}(x)$, for any valid set of share elements $T \subseteq [n] \times [L]$, there exists a subset $T' \subseteq T$ such that $\sum_{(i,j) \in T'} \mathbf{v}_{i,j} = x$, where $\mathbf{v}_i = (\mathbf{v}_{i,1}, \dots, \mathbf{v}_{i,L})$.

⁷ [Bon+18] only assumed a weaker property for their threshold FHE construction. However, this is a mistake introduced when merging the two works [JRS17] and [Bon+17] (and has been confirmed by the authors of [JRS17]).

Sharing Values in R_q . In our constructions, we share $\mathbf{x} \in R_q^r$, where $R_q = \mathbb{Z}_q[X]/f(X)$, instead of just in $\mathbb{Z}_q$. We do this coefficient-wise, by separately sharing each coefficient of the r polynomials in $\mathbf{x}$. Each party's share then lies in $(R_q^r)^L$, and the parties can perform R_q -linear operations on these shares.

Example Linear Secret Sharing Schemes. In Table 1, we detail a few example secret sharing schemes we consider. The schemes are for t -out-of- n access structures, where any $t + 1$ parties can reconstruct, and they all have strong $\{0, 1\}$ -reconstruction. In the table, we show two quantities $\tau_{\max}, \tau_{\min}$, which are relevant for choosing parameters in our constructions of Section 5 and we will refer to later. By $\tau_{\max}$ we denote the size of the smallest maximal invalid set of share elements, while $\tau_{\min}$ is the size of the largest minimal valid set of share elements.

Table 1. Example t -out-of- n linear secret sharing schemes with strong $\{0, 1\}$ -reconstruction. Details for the last row are omitted, due to their complexity.

Scheme	Sharing method	P_i 's share	L	$\tau_{\max}$	$\tau_{\min}$
Additive	$x = \sum_{i=1}^n x_i$	x_i	1	$n - 1$	n
Replicated	$x = \sum_{A, A =t} x_A$	$\{x_A\}_{i \notin A}$	$\binom{n-1}{t}$	$(n-t)(\binom{n}{t} - 1)$	$\binom{n}{t}$
Naive	$x = \sum_{i \in A} x_{A,i}, A = t + 1$	$\{x_{A,i}\}_{i \in A}$	$\binom{n-1}{t}$	$t \binom{n}{t+1}$	$t + 1$
Monotone Boolean formula for threshold fn. [Val84]			$O(n^{4.3})$	$O(n^{5.3})$	$O(n^{5.3})$

Additive Secret Sharing. In the $(n - 1)$ -out-of- n case, we use simple additive secret sharing, where x is split into random shares $x_1, \dots, x_n \in \mathbb{Z}_q$ such that $x = \sum_{i=1}^n x_i$. Every party receives exactly one share, hence $L = 1$, $\tau_{\max} = n - 1$ and $\tau_{\min} = n$.

Replicated Secret Sharing [ISN89]. To share x using replicated secret sharing (also called CNF sharing), first sample a set of additive shares $\{s_A\}_A$, over all size- t subsets $A \subset [n]$, such that $\sum_A s_A = x$. Then, party P_i 's share consists of every s_A where $i \notin A$. The share size is $L = \binom{n-1}{t}$.

A maximal invalid set contains all the copies of s_A for $A \neq A'$, for some A' . Since $n - t$ parties get A' , this gives $\tau_{\max} = nL - (n - t) = (n - t)(\binom{n}{t} - 1)$. On the other hand, a minimal valid set of share elements contains every share s_A , so $\tau_{\min} = \binom{n}{t}$.

Naive Threshold Secret Sharing. In the simplest form of threshold secret sharing, which can be seen as the dual of replicated secret sharing, the dealer

distributes a fresh sharing of x to each set S of size $t + 1$. There are $\binom{n}{t+1}$ such sets, but only $\binom{n-1}{t}$ of these contain party P_i , so $L = \binom{n-1}{t}$. It's easy to see that $\tau_{\max} = t \binom{n}{t+1}$ and $\tau_{\min} = t + 1$.

Threshold LSS From Monotone Boolean Formulae. An asymptotically more efficient approach is the construction of Benaloh and Leichter [BL90], which builds a linear secret scheme for $\mathbb{A}$ using any monotone Boolean formula for verifying membership of $\mathbb{A}$. A monotone Boolean formula is a circuit with AND/OR gates of fan-in 2 and fan-out 1, where the input wires may have multiple fan-out. The share size of party P_i equals the fan-out of the i -th input wire in the circuit.

Valiant [Val84] described a randomized construction of a monotone Boolean formula for threshold functions with size $O(n^{5.3})$. This leads to an average share size of $O(n^{4.3})$. In [HMP06], an improved circuit of size $O(n^{1+\sqrt{2}})$ was given, however, their circuit is not a formula, so cannot be used to build threshold LSS.

2.3 Learning With Errors

In the following, we recall the definitions of the decision (module) LWE problem [Reg05; LS15], formulated with a bounded uniform secret and noise. Let $R_q = \mathbb{Z}_q[X]/f(X)$ for some irreducible $f(X)$ of degree d . Further, we define $S_\beta = \{a \in R : \|a\|_\infty \leq \beta\}$ with $\beta \in \mathbb{N}$.

Definition 9 (M-LWE). Let $m, r, \beta, q \in \mathbb{N}$. The Module Learning With Errors problem M-LWE $_{q,m,r,\beta}$ is defined as follows. Given $\mathbf{A} \leftarrow U(R_q^{m \times r})$ and $\mathbf{t} \in R_q^m$. Decide whether $\mathbf{t} \leftarrow U(R_q^m)$ or if $\mathbf{t} = [\mathbf{A} | \mathbf{I}_m] \cdot \mathbf{s}$, where $\mathbf{s} \leftarrow U(S_\beta^{m+r})$.

The special case of $d = 1$, where the ring R is isomorphic to $\mathbb{Z}$, is simply denoted LWE (and is historically the one that has been introduced first).

We also define a computational variant of LWE, where no reduction modulo q is performed [Boo+18], which will be relevant in Section 6.

Definition 10 (I-LWE). Let $m, r \in \mathbb{N}$ and let χ_w, χ_e be two probability distributions over $\mathbb{Z}$. The Integer Learning With Errors problem I-LWE $_{m,r,\chi_w,\chi_e}$ is defined as follows. Given $\mathbf{W} \leftarrow \chi_w^{m \times r}$ and $\mathbf{t} = \mathbf{W}\mathbf{z} + \mathbf{e}$, where $\mathbf{z} \in \mathbb{Z}^r$ and $\mathbf{e} \leftarrow \chi_e^m$. Find $\mathbf{z}$. We call $(\mathbf{W}, \mathbf{t} = \mathbf{W}\mathbf{z} + \mathbf{e})$ an instance of the I-LWE distribution.

Theorem 1 ([Boo+18, Thm. 4.5]). Suppose that χ_w is τ_w -subgaussian and χ_e is τ_e -subgaussian. Let $(\mathbf{W}, \mathbf{t} = \mathbf{W}\mathbf{z} + \mathbf{e})$ be an instance of the I-LWE $_{m,r,\chi_w,\chi_e}$ distribution for some $\mathbf{z} \in \mathbb{Z}^r$. There exist constants $C_1, C_2 > 0$ such that for all $\nu \geq 1$ the least square method recovers $\mathbf{z}$ with probability $1 - \frac{1}{2r} - 2^{-\nu}$ if

$$m \geq 4 \frac{\tau_w^4}{\sigma_w^4} (C_1 r + C_2 \nu) \text{ and } m \geq 32 \frac{\tau_e^2}{\sigma_w^2} \log_2(2r).$$

3 Threshold Fully Homomorphic Encryption

In this section, we recall the definition of threshold fully homomorphic encryption schemes (ThFHE) and give different notions of robustness for threshold public key encryption, which model an adversary who may send incorrect or missing partial decryptions. We then define our notions of OW-CPA and IND-CPA security for ThFHE schemes.

3.1 Syntax and Basic Properties of Threshold FHE/PKE

We first recall the syntax of a fully homomorphic threshold public key encryption scheme. We implicitly assume that after **Setup**, all algorithms are given the public parameters as input. We omit the partial verification algorithm used in previous works (e.g., [BBH06]), which was only used to model stronger notions of robustness that also capture CCA attacks.

Definition 11 (ThFHE). A fully homomorphic threshold public key encryption scheme (ThFHE) for a message space $\mathcal{M}$ and circuits of depth κ is a tuple of PPT algorithms $\text{ThFHE} = (\text{Setup}, \text{Enc}, \text{Eval}, \text{PartDec}, \text{Combine})$ defined as follows:

- $\text{Setup}(1^\lambda, 1^\kappa, n, t) \rightarrow (\text{pp}, \text{pk}, \text{sk}_1, \dots, \text{sk}_n)$: On input the security parameter λ , a bound on the circuit depth κ , the number of parties n and a threshold value $t \in \{1, \dots, n-1\}$, the setup algorithm outputs the public parameters pp , a public key pk and a set of secret key shares $\text{sk}_1, \dots, \text{sk}_n$.
- $\text{Enc}(\text{pk}, m) \rightarrow \text{ct}$: On input the public key pk and a message $m \in \mathcal{M}$, the encryption algorithm outputs a ciphertext ct .
- $\text{Eval}(\text{pk}, C, \text{ct}_1, \dots, \text{ct}_k) \rightarrow \text{ct}$: On input the public key pk , a circuit $C: \mathcal{M}^k \rightarrow \mathcal{M}$ of depth at most κ and a set of ciphertexts $\text{ct}_1, \dots, \text{ct}_k$, the evaluation algorithm outputs a ciphertext ct .
- $\text{PartDec}(\text{sk}_i, \text{ct}) \rightarrow d_i$: On input a key share sk_i for some $i \in [n]$ and a ciphertext ct , the partial decryption algorithm outputs a partial decryption share d_i .
- $\text{Combine}(\{d_i\}_{i \in S}, \text{ct}) \rightarrow m'$: On input a set of decryption shares $\{d_i\}_{i \in S}$ and a ciphertext ct , where $S \subset [n]$ is of size at least $t+1$, the combining algorithm outputs a message $m' \in \mathcal{M} \cup \{\perp\}$.

The above can be seen as a generalization encompassing non-threshold and threshold PKE and FHE.

Definition 12 (ThPKE). A threshold public key encryption scheme (ThPKE) for a message space $\mathcal{M}$ is a ThFHE scheme, where $k = 1$ and the only allowed circuit $C: \mathcal{M} \rightarrow \mathcal{M}$ is the identity. In this case, we drop the trivial evaluation algorithm **Eval** and the parameter κ in the scheme's specifications.

Definition 13 (FHE). A fully homomorphic public key encryption scheme (FHE) for a message space $\mathcal{M}$ is a ThFHE scheme, where $n = 1$. In this case, we drop the parameters n and t in the scheme's specifications. To simplify notations, we merge **PartDec** and **Combine** into one single algorithm that we denote **Dec**. Hence, the algorithm **Dec** takes sk and ct as input and outputs $m' \in \{\mathcal{M} \cup \{\perp\}\}$.

We require compactness and correctness, whose definitions we recall in App. A.

In Section 4, we also need FHE schemes which are circuit private, defined below. To achieve this, we will rely on the construction of [DS16], which allows for LWE with polynomial modulus, since it can upgrade essentially any LWE-based FHE scheme to achieve circuit privacy without substantially increasing the parameters.⁸

Definition 14 (Circuit Privacy). *Let $s, \varepsilon > 0$. A ThFHE scheme with message space $\mathcal{M}$ and maximal circuit depth κ fulfills (s, ε) -circuit privacy if for every circuit C of depth at most κ it yields*

$$\text{cdist}_s \left(((\text{sk}_i)_{i \in n}, \text{Eval}(\text{pk}, C, \text{ct}_1, \dots, \text{ct}_k)), ((\text{sk}_i)_{i \in [n]}, \text{Enc}(\text{pk}, C(m_1, \dots, m_k))) \right) \leq \varepsilon,$$

where $m_i \in \mathcal{M}$ and $\text{ct}_i \leftarrow \text{Enc}(\text{pk}, m_i)$ for all $i \in [k]$ and for honestly generated keys $(\text{pk}, \text{sk}_1, \dots, \text{sk}_n)$.

3.2 Robustness

We now introduce two definitions of robustness for threshold public key encryption. We do not define these in the fully homomorphic case, where our construction assumes a passive adversary. We call the first one *weak chosen-ciphertext robustness* and the second *strong chosen-plaintext robustness*.

In the first case, it should be hard for an adversary, having access to all secret key shares, to provide one single ciphertext and two different set of decryption shares such that they combine to two different messages. Our definition is closely related to the notion of *consistency*, as for instance defined by [BBH06], with the difference that we do not allow the adversary to win by making the decryption output $\perp$. (This is unavoidable in our setting, since we do not have a separate PartVerify algorithm to verify validity of decryption shares.)

Definition 15 (Weak Chosen-Ciphertext Robustness). *We call a ThPKE scheme weakly chosen-ciphertext robust if for all λ, n, t and for all PPT adversaries $\mathcal{A}$ it yields*

$$\text{Adv}_{\text{ThPKE}}^{\text{w-cc-robust}}(\mathcal{A}) := \Pr[\text{Expt}_{\mathcal{A}, \text{ThPKE}}^{\text{w-cc-robust}}(1^\lambda, n, t) = 1] = \text{negl}(\lambda),$$

where $\text{Expt}_{\mathcal{A}, \text{ThPKE}}^{\text{w-cc-robust}}$ is the experiment specified in Figure 1.

In the second case, the adversary is given the secret key shares of the corrupted parties together with an honestly formed ciphertext. In order to win the experiment, they have to come up with partial decryption shares such that the combine algorithm, together with honestly generated partial decryption shares, outputs a different message (including the abort message $\perp$).

We note that for $t < n/2$, it's possible to transform *any* weakly chosen-ciphertext robust ThPKE scheme into one that guarantees strong chosen-plaintext

⁸ We could also use the construction of [Bou+16], however, it is restricted to evaluating log-depth circuits.

$\text{Expt}_{\mathcal{A}, \text{ThPKE}}^{\text{w-cc-robust}}(1^\lambda, n, t)$

```

1 : (pp, pk, sk1, ..., skn) ← Setup(1λ, n, t)
2 : (ct, S, S', {di}i∈S, {d'i}i∈S') ← A(pp, pk, {ski}i∈[n])
3 : m ← Combine({di}i∈S, ct)
4 : m' ← Combine({d'i}i∈S', ct)
5 : return m' ≠ m ∧ ⊥ ∉ {m, m'}

```

Fig. 1. Experiment for the weak chosen-ciphertext robustness of ThPKE schemes.

robustness. To do so, one simply lets Combine try all possible subsets of size $t+1$. As $t < n/2$, there exists a set of size $t+1$ composed of only honest partial decryption shares and hence, it successfully combines to a message.

Definition 16 (Strong Chosen-Plaintext Robustness). *A ThPKE scheme provides strong chosen-plaintext robustness if for all λ, n, t and for all PPT adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ it yields*

$$\text{Adv}_{\text{ThPKE}}^{\text{s-cp-robust}}(\mathcal{A}) := \Pr[\text{Expt}_{\mathcal{A}, \text{ThPKE}}^{\text{s-cp-robust}}(1^\lambda, n, t) = 1] = \text{negl}(\lambda),$$

where $\text{Expt}_{\mathcal{A}, \text{ThPKE}}^{\text{s-cp-robust}}$ is the experiment specified in Figure 2.

$\text{Expt}_{\mathcal{A}, \text{ThPKE}}^{\text{s-cp-robust}}(1^\lambda, n, t)$

```

1 : (pp, pk, sk1, ..., skn) ← Setup(1λ, n, t)
2 : (S, m) ← A1(pp, pk): S ⊂ [n] ∧ |S| ≤ t
3 : ct ← Enc(pk, m)
4 : dj ← PartDec(skj, ct), ∀ j ∈ [n] \ S
5 : {di}i∈S ← A2(pk, {ski}i∈S, {dj}j∉S, ct)
6 : m' ← Combine({di}i∈[n], ct)
7 : return m' ≠ m

```

Fig. 2. Experiment for strong chosen-plaintext robustness of ThPKE schemes.

3.3 One-Wayness

We now present our definition of OW-CPA security for ThFHE schemes.

The high level idea of the security game is the following. At the beginning, the adversary decides on the parties they want to corrupt and receives the corresponding secret key shares. We call this the static corruption setting. Then the adversary has access to three different oracles. The first, OEnc , allows them to obtain honestly generated, fresh ciphertexts on messages of their choice. Through

the second oracle, OChallEnc , the adversary obtains encryptions of unknown, randomly chosen messages, which we call the challenge messages and challenge ciphertexts. Finally, they can query up to ℓ times the last oracle, OPartDec , by inputting a circuit and a list of indices referring to previous encryption and challenge encryption queries, and receiving the corresponding partial decryption shares of all parties (after the evaluation algorithm has been applied). However, the partial decryption oracle aborts if for one of the challenge messages the conditional min-entropy has decreased more than an allowed amount ν , after having learned the circuit evaluation. Note that we do not condition the information-theoretical min-entropy on ChallCT as it uniquely defines ChallM . Implicitly, we assume that the entropy condition can be efficiently verified for the circuits input to OPartDec . One way to practically implement this, is to ask the adversary to input an algorithm which verifies the entropy condition when querying the oracle. We stress that in the transformation of Section 4 we only query circuits for which the entropy condition can be checked efficiently. To highlight the query bound ℓ and the entropy loss bound ν , we write (ℓ, ν) -OW-CPA.

Definition 17 ((ℓ, ν) -OW-CPA for ThFHE). *We call a ThFHE scheme (ℓ, ν) -OW-CPA secure for the security parameter λ , the circuit depth bound κ , the threshold parameters n, t , the query bound ℓ and the entropy bound ν , if for all PPT adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$*

$$\text{Adv}_{\text{ThFHE}}^{(\ell, \nu)\text{-OW-CPA}}(\mathcal{A}) := \Pr[\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{(\ell, \nu)\text{-OW-CPA}}(1^\lambda, 1^\kappa, n, t) = 1] = \text{negl}(\lambda),$$

where $\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{(\ell, \nu)\text{-OW-CPA}}$ is the experiment in Fig. 3 with $\text{ctr} = 0$, $\text{idx} = 0$ and $\text{L} = \emptyset$ at the beginning.

Definition 18 (ℓ -OW-CPA for ThPKE). *We call a ThPKE scheme ℓ -OW-CPA secure for the security parameter λ , the threshold parameters n, t and the query bound ℓ , if it is $(\ell, 0)$ -OW-CPA secure as ThFHE scheme, where $k = 1$ and the only allowed circuit C is the identity. In this case, the OPartDec oracle from Figure 3 only replies to ciphertexts that have been output by OEnc and aborts if the ciphertext has been output by OChallEnc , as the entropy is zero for every challenge ciphertext and hence never passes the entropy check. For a PPT adversary $\mathcal{A}$, we denote their advantage by $\text{Adv}_{\text{ThPKE}}^{\ell\text{-OW-CPA}}(\mathcal{A})$.*

3.4 Indistinguishability

In the following, we present two definitions of IND-CPA security for ThFHE, both with an apriori upper bound ℓ on the number of partial decryption queries. We call the first **selective- ℓ -IND-CPA**. It can be seen as a slightly weaker version of the original game-based security notion in [JRS17, Def. 14]. Both definitions have a selective nature. The only difference is that in our definition, challenge messages and circuit queries are sent together, whereas in [JRS17] the messages and circuit queries are sent in two different phases. We call the second version we

$\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{(\ell, \nu)\text{-OW-CPA}}(1^\lambda, 1^\kappa, n, t)$

```

1 : (pp, pk, sk1, ..., skn) ← Setup(1λ, 1κ, n, t)
2 : S ← A1(pp, pk): S ⊂ [n] ∧ |S| ≤ t
3 : (m', j) ← A2OEnc, OChallEnc, OPartDec(pk, {ski}i∈S)
4 : (bj, mj, ctj) := L[j]
5 : return mj = m' ∧ bj = 1

```

OEnc(m) <pre> 1 : if m ∉ M then return ⊥ 2 : idx = idx + 1 3 : ct ← Enc(pk, m) 4 : CT = CT ∪ {ct} 5 : L[idx] := {(0, m, ct)} 6 : return ct </pre>	OChallEnc() <pre> 1 : idx = idx + 1 2 : m ← M 3 : ChallM = ChallM ∪ {m} 4 : ct ← Enc(pk, m) 5 : ChallCT = ChallCT ∪ {ct} 6 : L[idx] := {(1, m, ct)} 7 : return ct </pre>
--	---

OPartDec(C, ℓ₁, ..., ℓ_k)

```

1 : ctr = ctr + 1
2 : if ctr > ℓ then return ⊥
3 : if ∃j ∈ [k]: ℓj > |L| then return ⊥
4 : if depth(C) > κ then return ⊥
5 : (bj, mj, ctj) := L[ℓj], j ∈ [k]
6 : ct ← Eval(pk, C, ct1, ..., ctk)
7 : di ← PartDec(ski, ct), i ∈ [n]
8 : d = (di)i∈[n]
9 : for m ∈ ChallM
10 :   if  $\tilde{H}_\infty(m | \mathbf{E} \cup \{(C, C(m_1, \dots, m_k))\}) < \tilde{H}_\infty(m) - \nu$  then
11 :     return ⊥
12 :   E = E ∪ {(C, C(m1, ..., mk))}
13 :   PartD = PartD ∪ {d}
14 : return d

```

Fig. 3. Experiment and oracles for (ℓ, ν) -OW-CPA security of ThFHE schemes.

introduce **adaptive- ℓ -IND-CPA**. It is significantly stronger, as the adversary can now adaptively query the (challenge) encryption and partial decryption oracles. For completeness, we also state a much weaker notion of indistinguishability in A.2 which appeared in the first version of this paper [BS23a]. Our first transformation in Section 4.1 applying to ThPKE fulfills the stronger adaptive notion, whereas the second transformation in Section 4.2 applying to ThFHE only leads to a scheme fulfilling the weaker selective notion. As explained in the introduction, the published version [BS23e] and earlier versions of this paper [BS23c] wrongly claimed that the resulting scheme even fulfills the adaptive notion.

Selective Indistinguishability. After having received the secret key shares of the statically corrupted parties, the adversary outputs a list of length ℓ composed of message pairs and circuits. Upon receiving the corresponding challenge ciphertexts and partial decryption shares, the adversary then outputs a guess.

Definition 19 (selective- ℓ -IND-CPA for ThFHE). *A ThFHE scheme is said to fulfill selective- ℓ -IND-CPA security for the security parameter λ , the circuit depth bound κ , the threshold parameters n, t and the query bound ℓ , if for all PPT adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3)$*

$$\text{Adv}_{\text{ThFHE}}^{\text{selective-}\ell\text{-IND-CPA}}(\mathcal{A}) := \left| \Pr[\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{selective-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t) = 1] - \frac{1}{2} \right| = \text{negl}(\lambda),$$

where $\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{selective-}\ell\text{-IND-CPA}}$ is the experiment in Fig. 4.

Adaptive Indistinguishability. As for the OW-CPA security, we allow for static corruptions and access to three different oracles. The first, **OEnc**, is the same as in the OW-CPA game. To the second oracle, **OChallEnc**, the adversary inputs two messages and obtains the encryption of one of it. Finally, they can again query up to ℓ times **OPartDec**, by inputting a circuit and a list of indices and receiving the corresponding partial decryption shares of all parties. This time, the partial decryption oracle aborts if the circuit evaluates to different values on the corresponding input messages to the **OChallEnc** oracle.

Definition 20 (adaptive- ℓ -IND-CPA for ThFHE). *A ThFHE scheme is said to fulfill adaptive- ℓ -IND-CPA security for the security parameter λ , the circuit depth bound κ , the threshold parameters n, t and the query bound ℓ , if for all PPT adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$*

$$\text{Adv}_{\text{ThFHE}}^{\text{adaptive-}\ell\text{-IND-CPA}}(\mathcal{A}) := \left| \Pr[\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{adaptive-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t) = 1] - \frac{1}{2} \right| = \text{negl}(\lambda),$$

where $\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{adaptive-}\ell\text{-IND-CPA}}$ is the experiment in Fig. 5 with $\text{ctr} = 0$, $\text{idx} = 0$ and $L = \emptyset$ at the beginning.

$\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{selective-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t)$

```

1 :  (pp, pk, sk1, ..., skn) ← Setup(1λ, 1κ, n, t)
2 :  S ← A1(pp, pk) : S ⊂ [n] ∧ |S| ≤ t
3 :  (m̄i(0), m̄i(1), Ci)i∈[ℓ] ← A2(pk, {ski}i∈S)
4 :  b ← {0, 1}
5 :  for i ∈ [ℓ] :
6 :    if depth(Ci) > κ then return ⊥
7 :    if Ci(m̄i(0)) ≠ Ci(m̄i(1)) then return ⊥
8 :    parse m̄i(b) := (mi1(b), ..., mik(b))
9 :    for j ∈ [k] :
10 :      ctij ← Enc(pk, mij(b))
11 :      cṫi := (ctij)j∈[k]
12 :      ĉi ← Eval(pk, Ci, cṫi)
13 :      for j ∈ [n] :
14 :        dij ← PartDec(skj, ĉi)
15 :      d̄i := (dij)j∈[n]
16 :      b' ← A3((cṫi, d̄i)i∈[ℓ])
17 :  return b = b'
```

Fig. 4. Experiment for selective- ℓ -IND-CPA security of ThFHE schemes.

$\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{adaptive-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t)$	
1 : $(\text{pp}, \text{pk}, \text{sk}_1, \dots, \text{sk}_n) \leftarrow \text{Setup}(1^\lambda, 1^\kappa, n, t)$ 2 : $S \leftarrow \mathcal{A}_1(\text{pp}, \text{pk}) : S \subset [n] \wedge S \leq t$ 3 : $b \leftarrow \{0, 1\}$ 4 : $b' \leftarrow \mathcal{A}_2^{\text{OEnc}, \text{OChallEnc}, \text{OPartDec}}(\text{pk}, \{\text{sk}_i\}_{i \in S})$ 5 : return $b = b'$	
$\text{OEnc}(m)$ 1 : if $m \notin \mathcal{M}$ then return $\perp$ 2 : $\text{idx} = \text{idx} + 1$ 3 : $\text{ct} \leftarrow \text{Enc}(\text{pk}, m)$ 4 : $\text{L}[\text{idx}] := \{(m, m, \text{ct})\}$ 5 : return ct	$\text{OChallEnc}(m^{(0)}, m^{(1)})$ 1 : if $(m^{(0)}, m^{(1)}) \notin \mathcal{M} \times \mathcal{M}$ then return $\perp$ 2 : $\text{idx} = \text{idx} + 1$ 3 : $\text{ct}_b \leftarrow \text{Enc}(\text{pk}, m^{(b)})$ 4 : $\text{L}[\text{idx}] := \{(m^{(0)}, m^{(1)}, \text{ct}_b)\}$ 5 : return ct_b
$\text{OPartDec}(C, \iota_1, \dots, \iota_k)$	
1 : $\text{ctr} = \text{ctr} + 1$ 2 : if $\text{ctr} > \ell$ then return $\perp$ 3 : if $\exists j \in [k] : \iota_j > \text{L} $ then return $\perp$ 4 : if $\text{depth}(C) > \kappa$ then return $\perp$ 5 : $(m_j^{(0)}, m_j^{(1)}, \text{ct}_j) := \text{L}[\iota_j], \quad j \in [k]$ 6 : if $C(m_1^{(0)}, \dots, m_k^{(0)}) \neq C(m_1^{(1)}, \dots, m_k^{(1)})$ then return $\perp$ 7 : $\text{ct} \leftarrow \text{Eval}(\text{pk}, C, \text{ct}_1, \dots, \text{ct}_k)$ 8 : $d_i \leftarrow \text{PartDec}(\text{sk}_i, \text{ct}), \quad i \in [n]$ 9 : return $(d_i)_{i \in [n]}$	

Fig. 5. Experiment and oracles for adaptive- ℓ -IND-CPA security of ThFHE schemes.

As Section 4.1 only applies to ThPKE we state for completeness the definition of adaptive- ℓ -IND-CPA security in this special case.

Definition 21 (adaptive- ℓ -IND-CPA for ThPKE). *We call a ThPKE scheme adaptive- ℓ -IND-CPA secure for the security parameter λ , the threshold parameters n, t and the query bound ℓ , if it is adaptive- ℓ -IND-CPA secure as ThFHE scheme, where $k = 1$ and the only allowed circuit C is the identity. In this case, the OPartDec oracle from Figure 5 only replies to ciphertexts that have been output by OEnc and aborts if the ciphertext has been output by OChallEnc. For a PPT adversary $\mathcal{A}$, we denote their advantage by $\text{Adv}_{\text{ThPKE}}^{\text{adaptive-}\ell\text{-IND-CPA}}(\mathcal{A})$.*

4 From One-Wayness to Indistinguishability

4.1 Transformation for Weakly Robust Threshold Decryption

A tight reduction from OW-CPA security to IND-CPA security for standard PKE schemes in the random oracle model (ROM) was provided in [HHK17, Sec. 3.4]. In the following, we adapt the transformation to the threshold setting and show how a small modification allows to obtain a weakly chosen-ciphertext robust threshold scheme as in Definition 15. The concrete flavor of indistinguishability obtained through this transformation is the adaptive- ℓ -IND-CPA security, cf. Definition 21.

The construction. The transformation is parameterized by $\delta \in \mathbb{N}$ which allows for a trade-off between the security loss of the reduction and the compactness of ciphertexts. Given $\text{ThPKE} = (\text{Setup}, \text{Enc}, \text{PartDec}, \text{Combine})$ with message space $\mathcal{M}$ being OW-CPA secure, we define $\text{ThPKE}' = (\text{Setup}', \text{Enc}', \text{PartDec}', \text{Combine}')$ with message space an abelian group $(\mathcal{M}', +)$, which fulfills IND-CPA security, as follows. Let $F: \mathcal{M}^\delta \rightarrow \mathcal{M}'$ and $G: \mathcal{M}^\delta \rightarrow \{0, 1\}^{2\lambda}$ be two random oracles.

Setup': On input $(1^\lambda, n, t)$, it outputs $(\text{pp}, \text{pk}, \text{sk}_1, \dots, \text{sk}_n) \leftarrow \text{Setup}(1^\lambda, n, t)$.
Enc': On input (pk, m) with $m \in \mathcal{M}'$, it samples $\mathbf{x} := (x_1, \dots, x_\delta) \leftarrow U(\mathcal{M}^\delta)$ and sets $c_0 = m + F(\mathbf{x})$ and $c_{\delta+1} = G(\mathbf{x})$. Then, it computes $c_j \leftarrow \text{Enc}(\text{pk}, x_j)$ for $j \in [\delta]$ and outputs $\text{ct} := (c_0, \dots, c_{\delta+1})$.
PartDec': On input (sk_i, ct) for some $i \in [n]$, it computes $d_{ij} \leftarrow \text{PartDec}(\text{sk}_i, c_j)$ for all $j \in [\delta]$ and outputs $\mathbf{d}_i := (d_{ij})_{j \in [\delta]}$.
Combine': On input $((\mathbf{d}_i)_{i \in S}, \text{ct})$ with $\text{ct} = (c_j)_{0 \leq j \leq \delta+1}$ and $\mathbf{d}_i = (d_{ij})_{j \in [\delta]}$, it computes $x'_j \leftarrow \text{Combine}(\{d_{ij}\}_{i \in S}, c_j)$ for $j \in [\delta]$, sets $\mathbf{x}' = (x'_1, \dots, x'_\delta)$ and computes $m' := c_0 - F(\mathbf{x}')$. If $c_{\delta+1} = G(\mathbf{x}')$ it outputs m' . Else, it outputs $\perp$.

Ciphertext expansion. The ratio between the bit size of the plaintext and the ciphertext is given by

$$\frac{|\text{ct}|}{|m|} = \frac{|m| + \delta \cdot |c| + 2\lambda}{|m|},$$

where c is a ciphertext coming from ThPKE. We can see that with larger δ the ciphertext expansion gets worse.

We prove the decryption correctness of the resulting scheme in Appendix B.1.

Lemma 7 (Weak Chosen-Ciphertext Robustness). *The scheme ThPKE' is weakly chosen-ciphertext robust. More precisely, if there is a PPT adversary $\mathcal{A}$ such that $\text{Adv}_{\text{ThPKE}'}^{\text{w-cc-robust}}(\mathcal{A}) \geq \varepsilon$ for some $\varepsilon > 0$, then there exists a PPT adversary $\mathcal{B}$ breaking collision resistance of the random oracle G with probability $\geq \varepsilon$.*

Proof. Fix λ, n and t . We show that if there exists a PPT adversary $\mathcal{A}$ that has advantage ε in the experiment defined in Figure 1, then there exists a PPT adversary $\mathcal{B}$ that finds a collision for the random oracle G with the same probability ε . Let $\mathcal{B}$ play the role of the challenger in the weak robustness game, running the Setup' algorithm on $(1^\lambda, n, t)$ and forwarding $(\text{pp}, \text{pk}, \{\text{sk}_i\}_{i \in [n]})$ to $\mathcal{A}$. Assume that $\mathcal{A}$ wins the weak robustness game by outputting two sets of decryption shares $\{\mathbf{d}_i\}_{i \in S}$ and $\{\mathbf{d}'_i\}_{i \in S'}$ such that $\text{Combine}'(\{\mathbf{d}_i\}_{i \in S}, \text{ct}) \rightarrow m \neq m' \leftarrow \text{Combine}'(\{\mathbf{d}'_i\}_{i \in S'}, \text{ct})$ for the same ciphertext $\text{ct} = (c_i)_{0 \leq i \leq \delta+1}$, and neither m nor m' equals $\perp$. Let $\mathbf{x}, \mathbf{x}'$ denote the vectors recovered during the combining procedure. As $c_0 = m + F(\mathbf{x}) = m' + F(\mathbf{x}')$, $m \neq m'$ and F is deterministic, we can deduce that $\mathbf{x} \neq \mathbf{x}'$. This implies that $G(\mathbf{x}) = c_{\delta+1} = G(\mathbf{x}')$ for distinct $\mathbf{x} \neq \mathbf{x}'$ and hence $\mathcal{B}$ has found a collision in G . $\square$

Theorem 2 (Security). *Let $\delta, \ell \in \mathbb{N}$. If ThPKE is $(\ell\delta)$ -OW-CPA secure, then ThPKE' fulfills adaptive- ℓ -IND-CPA security in the ROM. More precisely, for any adaptive- ℓ -IND-CPA adversary $\mathcal{A}$ that does at most q_F queries to the random oracle F and q_c queries to the oracle $\text{OChallEnc}'$, there exists an $(\ell\delta)$ -OW-CPA adversary $\mathcal{B}$ with*

$$\text{Adv}_{\text{ThPKE}'}^{\text{adaptive-}\ell\text{-IND-CPA}}(\mathcal{A}) \leq q_c \cdot q_F^{1/\delta} \cdot \text{Adv}_{\text{ThPKE}}^{(\ell\delta)\text{-OW-CPA}}(\mathcal{B}).$$

Note that the number of queries to G doesn't impact the tightness of the reduction as the output $G(\mathbf{x})$ is completely independent of $F(\mathbf{x})$ for any $\mathbf{x} \in \mathcal{M}^\delta$. Moreover, there is no entropy bound involved as we are in the standard ThPKE setting (Def. 18).

Proof. The proof closely follows the original proof in [HHK17, Thm. 3.7]. The main modifications compared to the original proof are that $\mathcal{A}$ can make multiple queries to OChallEnc (leading to a security loss of q_c), can further query up to ℓ partial decryption outputs to some oracle OPartDec during the game and that we added a second random oracle G to obtain weak robustness.

Let $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ be a PPT adversary against the adaptive- ℓ -IND-CPA security of ThPKE' . We consider two games G_0 and G_1 as described in Figure 6, where we specify the security game, the queries to the random oracles F and G and to the oracle $\text{OChallEnc}'$ from Def. 21. We omit the specification of OEnc' and $\text{OPartDec}'$ as they follow directly from the construction of the scheme and the security definition. The lists $\mathcal{L}_F$ and $\mathcal{L}_G$ are initialized as empty sets and the counters ctr and idx are set to 0 at the beginning. Both games only differ in the way how queries to F are handled.

Games G_0 and G_1		$\text{OChallEnc}'(m^{(0)}, m^{(1)})$	
1 :	$(\text{pp}, \text{pk}, \text{sk}_1, \dots, \text{sk}_n) \leftarrow \text{Setup}(1^\lambda, n, t)$	1 :	$\text{idx} = \text{idx} + 1$
2 :	$S \leftarrow \mathcal{A}_1(\text{pp}, \text{pk}) : S \subset [n] \wedge S \leq t$	2 :	choose next unused $\mathbf{x} \in \text{ChallX}$
3 :	$\text{ChallX} \leftarrow U((\mathcal{M}^\delta)^{q_c})$	3 :	$c_0 = m_b + F(\mathbf{x})$
4 :	$b' \leftarrow \mathcal{A}_2^{\text{OEnc}, \text{OChallEnc}, \text{OPartDec}, F, G}(\text{pk}, \{\text{sk}_i\}_{i \in S})$	4 :	$c_j \leftarrow \text{Enc}(\text{pk}, x_j) : j \in [\delta]$
5 :	return b'	5 :	$c_{\delta+1} = G(\mathbf{x})$
		6 :	$\text{ct}_b = (c_0, \dots, c_{\delta+1})$
		7 :	$L[\text{idx}] := \{(m^{(0)}, m^{(1)}, \text{ct}_b)\}$
		8 :	return ct_b
$F(\mathbf{x})$		$G(\mathbf{x})$	
1 :	if $\exists r : (\mathbf{x}, r) \in \mathcal{L}_F$	1 :	if $\exists r : (\mathbf{x}, r) \in \mathcal{L}_G$
2 :	then return r	2 :	then return r
3 :	if $\mathbf{x} \in \text{ChallX}$	3 :	$r \leftarrow U(\{0, 1\}^{2\lambda})$
4 :	$\text{flag} = \text{true}$	4 :	$\mathcal{L}_G := \mathcal{L}_G \cup \{(\mathbf{x}, r)\}$
5 :	then return $\perp$	5 :	return r
6 :	$r \leftarrow U(\{0, 1\}^\lambda)$		
7 :	$\mathcal{L}_F := \mathcal{L}_F \cup \{(\mathbf{x}, r)\}$		
8 :	return r		

Fig. 6. Games G_0 and G_1 for the proof of Theorem 2.

Game G_0 . Note that Game G_0 is exactly the original adaptive- ℓ -IND-CPA game (as in Def. 21) and hence $\text{Adv}_{\text{ThPKE}'}^{\text{adaptive-}\ell\text{-IND-CPA}}(\mathcal{A}) = |\Pr[G_0(\mathcal{A}) = 1] - 1/2|$.

Game G_1 . The only modification between game G_0 and G_1 is that we added line 3–5 in the specification of $\mathbf{F}$. More precisely, $\mathbf{F}$ raises a flag and aborts if it is queried by one of the vectors in $\mathbf{ChallX}$ that are used for the challenge ciphertexts issued from $\mathbf{OChallEnc}'$. Hence, $|\Pr[G_0(\mathcal{A}) = 1] - \Pr[G_1(\mathcal{A}) = 1]| \leq \Pr[\text{flag}]$. Now, as $\mathbf{F}$ aborts when queried on $\mathbf{x} \in \mathbf{ChallX}$, the view of $\mathcal{A}$ is independent of the bit b chosen in the experiment and defining $\mathbf{OChallEnc}'$. This implies that $\Pr[G_1(\mathcal{A}) = 1] = 1/2$, leading to $\text{Adv}_{\text{ThPKE}'}^{\text{adaptive-}\ell\text{-IND-CPA}}(\mathcal{A}) \leq \Pr[\text{flag}]$. The only thing left to do is to bound the latter probability. A direct adaptation of Lemma 3.8 in [HHK17], together with the union bound, bounds this probability above by $q_c \cdot q_F^{1/\delta} \cdot \text{Adv}_{\text{ThPKE}}^{(\ell\delta)\text{-OW-CPA}}(\mathcal{A})$. Here, the adversary $\mathcal{A}$ is embedded in $\mathcal{B}$'s own $(\ell\delta)$ -OW-CPA security game and hence $\mathcal{B}$ takes care of simulating the random oracles $\mathbf{F}$ and $\mathbf{G}$ as well as the oracles $\mathbf{OEnc}'$, $\mathbf{OChallEnc}'$ and $\mathbf{OPartDec}'$. The latter is done by querying their own partial decryption oracle $\mathbf{OPartDec}$. Note that the increase from ℓ to $\ell\delta$ comes from the fact that $\mathcal{B}$ must do δ queries to $\mathbf{OPartDec}$ for every query to $\mathbf{OPartDec}'$ by $\mathcal{A}$. $\square$

4.2 For Fully Homomorphic Threshold Decryption

Whereas the reduction from above is simple and tight, it has the disadvantage of needing the random oracle $\mathbf{F}$ to mask the message m . When considering not only threshold PKE, but more generally threshold FHE, we need to make sure that we can homomorphically evaluate ciphertexts. The use of the random oracle $\mathbf{F}$ when computing $c_0 = m + \mathbf{F}(\mathbf{x})$ makes such an evaluation impossible, as there is no finite circuit description of the random oracle $\mathbf{F}$. We thus need another transformation which allows for homomorphic evaluation of ciphertexts.

In the following, we describe a generic way of transforming a OW-CPA secure ThFHE scheme into an IND-CPA secure one in the standard model, via hardcore bits. The concrete flavor of indistinguishability obtained through this transformation is the selective- ℓ -IND-CPA security, cf. Definition 19.

The construction. The transformation is parameterized by $\delta, \gamma \in \mathbb{N}$. Given $\text{ThFHE} = (\text{Setup}, \text{Enc}, \text{Eval}, \text{PartDec}, \text{Combine})$ with message space $\mathcal{M} = \{0, 1\}^\gamma$ being OW-CPA secure, we define $\text{ThFHE}' = (\text{Setup}', \text{Enc}', \text{Eval}', \text{PartDec}', \text{Combine}')$ with message space $\mathcal{M}' = \{0, 1\}^\delta$, which we show to fulfill IND-CPA security, as follows.

Setup': On input $(1^\lambda, 1^\kappa, n, t)$, it outputs $(\text{pp}, \text{pk}, \text{sk}_1, \dots, \text{sk}_n) \leftarrow \text{Setup}(1^\lambda, 1^\kappa, n, t)$.
Enc': On input (pk, m) with $m = (m_j)_{j \in [\delta]} \in \mathcal{M}'$, it samples $x \leftarrow U(\mathcal{M})$ and computes $c_0 \leftarrow \text{Enc}(\text{pk}, x)$. For $j \in [\delta]$, it samples $s_j \leftarrow U(\mathcal{M})$ and computes $c_j = \langle x, s_j \rangle + m_j \bmod 2$. It outputs $\text{ct} = (c_0, s_1, \dots, s_\delta, c_1, \dots, c_\delta)$.
Eval': On input $I := (\text{pk}, C, \text{ct}_1, \dots, \text{ct}_k)$, where $\text{ct}_i = (c_{i0}, s_{i1}, \dots, s_{i\delta}, c_{i1}, \dots, c_{i\delta})$ such that $c_{i0} \leftarrow \text{Enc}(\text{pk}, x_i)$ for $i \in [k]$ and $C: (\mathcal{M}')^k \rightarrow \mathcal{M}'$, it first defines a circuit $\tilde{C}: (\mathcal{M})^k \rightarrow \mathcal{M}$ as follows:

- $\tilde{C}$ takes as input $(x_1, \dots, x_k)$ and has the information I hard-coded
- It computes $m_{ij} = c_{ij} + \langle x_i, s_{ij} \rangle \bmod 2$, for $j \in [\delta]$ and $i \in [k]$
- It outputs $C(m_1, \dots, m_k)$, where $m_i = (m_{ij})_{j \in [\delta]}$

It then outputs $\text{ct}' = \text{Eval}(\text{pk}, \tilde{C}, c_{10}, \dots, c_{k0})$.

PartDec': On input $(\text{sk}_i, \text{ct}')$, it outputs $d_i = \text{PartDec}(\text{sk}_i, \text{ct}')$.

Combine': On input $(\{d_i\}_{i \in S}, \text{ct}')$, it outputs $m = \text{Combine}(\{d_i\}_{i \in S}, \text{ct}')$.

Ciphertext expansion. The ratio between the bit size of the plaintext and the ciphertext is give by

$$\frac{|\text{ct}|}{|m|} = \frac{|c_0| + \delta(\gamma + 1)}{\delta},$$

where c_0 is the OW-CPA ciphertext encrypting γ bits coming from ThFHE. We can see that with larger δ the ciphertext expansion gets better.

We prove compactness and decryption correctness in Appendix B.2.

Remark 1. One way to reduce the size of the ciphertext to $|c_0| + \gamma + \delta$ (and hence to improve the ciphertext expansion) is to replace the δ random seeds $s_1, \dots, s_\delta$ by one single seed and a random oracle F . More precisely, one could define $s_j := F(r, j)$ for a random seed $r \leftarrow U(\mathcal{M})$ and $j \in [\delta]$. As a result, the transformation wouldn't be in the standard, but in the random oracle model. As the random oracle is only used to derive the seeds, not when masking the message, this transformation still applies to the threshold FHE setting.

Remark 2. Note that the reduction in the standard model restricted to ThPKE, in contrast to the one from Section 4.1, doesn't satisfy weak robustness (Def. 15).

Theorem 3 (Security). Fix $\ell, k, \delta, \gamma, \lambda, s \in \mathbb{N}$ and $\varepsilon > 0$, where k denotes the number of variables that every circuit takes as input. Let ThFHE be an $(\ell(k+1), \delta)$ -OW-CPA secure scheme with $\mathcal{M} = \{0, 1\}^\gamma$, such that any adversary $\mathcal{B}$ of circuit size s has advantage $\text{Adv}_{\text{ThFHE}}^{((k+1)\ell, \delta)\text{-OW-CPA}}(\mathcal{B}) \leq 2^{-\lambda}$, where $\lambda \geq 3 \log_2(1/\varepsilon) + \log_2(2\gamma) + \delta$. Further, we assume that ThFHE fulfills (s', ε') -circuit privacy, where $s' = O(s\gamma^3\varepsilon^{-4})$ ⁹ and $\varepsilon' = 5k\delta\varepsilon$. Then, ThFHE' is selective- ℓ -IND-CPA secure with $\mathcal{M}' = \{0, 1\}^\delta$; concretely, for any adversary $\mathcal{A}$ of circuit size s' it yields

$$\text{Adv}_{\text{ThFHE}'}^{\text{selective-}\ell\text{-IND-CPA}}(\mathcal{A}) \leq \frac{25(k+1)\ell\delta\varepsilon + 1}{2}.$$

⁹ The hidden constant in the $O(\cdot)$ notation is the same as that in the proof of Lemma 1, which can be derived from the Goldreich-Levin theorem.

Choosing the Parameters. To ensure a small enough advantage, since k, ℓ, δ are relatively small, it suffices to choose a small enough ε , which we denote $\varepsilon = 2^{-\lambda'}$. We then require $\lambda = \delta + 3\lambda' + \log_2(2\gamma)$, which determines the required security level of the original OW-CPA scheme. There's therefore a tradeoff between the increased security requirement and the value δ , which improves ciphertext expansion. For instance, if $\lambda' = 128$ then by choosing $\delta = 118, \gamma = 512$, we can pack 118 message bits into each FHE ciphertext, which must encrypt 512 actual bits using the OW-CPA scheme. In this case, to achieve security according to the reduction, the parameters of the OW-CPA scheme would need to be chosen for $\lambda = 512$ -bit security. We note that this way of setting parameters may be overly conservative, since our reduction is not tight — unlike with the number of queries ℓ and the matching attack (Section 6.2), we are not aware of any weaknesses from choosing smaller values of λ .

Proof. Recall that we are given an OW-CPA secure threshold decryption scheme $\text{ThFHE} = (\text{Setup}, \text{Enc}, \text{Eval}, \text{PartDec}, \text{Combine})$ with message space $\mathcal{M} = \{0, 1\}^\gamma$ and we want to construct a new threshold scheme $\text{ThFHE}' = (\text{Setup}', \text{Enc}', \text{Eval}', \text{PartDec}', \text{Combine}')$ with message space $\mathcal{M}' = \{0, 1\}^\delta$, which fulfills the selective- ℓ -IND-CPA security. In the selective- ℓ -IND-CPA security game (Def. 19), the adversary, after having received secret key shares of the corrupted parties, sends a list of challenge messages and circuits, for which they receive as answer the corresponding challenge ciphertexts and partial decryption shares. In the following, we define a sequence of games which modify how the answers to the adversary are computed. The first game consists of the selective- ℓ -IND-CPA security game, where $b = 1$. The last game consists of the selective- ℓ -IND-CPA security game, where $b = 0$.

Game₀ : Let $(\vec{m}_i^{(0)}, \vec{m}_i^{(1)}, C_i)_{i \in [\ell]}$ be the ℓ message vector pairs (each vector of dimension k) and ℓ circuits output by the adversary.

Encryption Queries: Every of the ℓ message vector pairs is composed of k simple message pairs. For every of those $k \cdot \ell$ message pairs $m^{(0)}, m^{(1)} \in \{0, 1\}^\delta = \mathcal{M}'$, sample $x, s_1, \dots, s_\delta \leftarrow U(\mathcal{M})$, compute $c_0 \leftarrow \text{Enc}(\text{pk}, x)$, and set $c_i = \langle x, s_i \rangle + m_i^{(1)}$ for all $i \in [\delta]$. Output $\text{ct} = (c_0, s_1, \dots, s_\delta, c_1, \dots, c_\delta)$.

Partial Decryption Queries: For each of the ℓ circuits C and corresponding challenge ciphertexts $\text{ct}_1, \dots, \text{ct}_k$ (computed above), compute $\text{ct} \leftarrow \text{Eval}'(\text{pk}, C, \text{ct}_1, \dots, \text{ct}_k)$ (by internally calling Eval on associated circuit $\tilde{C}$) and then $d_i \leftarrow \text{PartDec}'(\text{sk}_i, \text{ct})$ (by internally calling PartDec) for all $i \in [n]$. Output $\mathbf{d} = (d_i)_{i \in [n]}$.

Game₁ :

Encryption Queries: as in Game₀

Partial Decryption Queries: For each of the ℓ circuits C and corresponding challenge messages $m_1^{(1)}, \dots, m_k^{(1)}$, define the constant circuit $\tilde{C}$ which, on any input simply outputs $C(m_1^{(1)}, \dots, m_k^{(1)})$. First compute $\text{ct} \leftarrow \text{Enc}(\text{pk}, \tilde{C}(x_1^{(1)}, \dots, x_k^{(1)}))$ (on arbitrary input $x_i^{(1)}$) and then $d_i \leftarrow \text{PartDec}(\text{sk}_i, \text{ct})$ for all $i \in [n]$. Output $\mathbf{d} = (d_i)_{i \in [n]}$.

Game₂ :

Encryption Queries: For each of the $k \cdot \ell$ message pairs $m^{(0)}, m^{(1)} \in \{0, 1\}^\delta = \mathcal{M}'$, sample $x, s_1, \dots, s_\delta \leftarrow U(\mathcal{M})$ and compute $c_0 \leftarrow \text{Enc}(\text{pk}, x)$. Further, sample $r_1, \dots, r_\delta \leftarrow U(\{0, 1\})$ and set $c_i = r_i + m_i^{(1)} \bmod 2$ for all $i \in [\delta]$. Output $\text{ct} = (c_0, s_1, \dots, s_\delta, c_1, \dots, c_\delta)$.

Partial Decryption Queries: as in Game₁

Game₃ :

Encryption Queries: For each of the $k \cdot \ell$ message pairs $m^{(0)}, m^{(1)} \in \{0, 1\}^\delta = \mathcal{M}'$, sample $x, s_1, \dots, s_\delta \leftarrow U(\mathcal{M})$ and compute $c_0 \leftarrow \text{Enc}(\text{pk}, x)$. Further, sample $r_1, \dots, r_\delta \leftarrow U(\{0, 1\})$ and set $c_i = r_i + m_i^{(0)} \bmod 2$ for all $i \in [\delta]$. Output $\text{ct} = (c_0, s_1, \dots, s_\delta, c_1, \dots, c_\delta)$.

Partial Decryption Queries: as in Game₂

Game₄ :

Encryption Queries: For each of the $k \cdot \ell$ message pairs $m^{(0)}, m^{(1)} \in \{0, 1\}^\delta = \mathcal{M}'$, sample $x, s_1, \dots, s_\delta \leftarrow U(\mathcal{M})$, compute $c_0 \leftarrow \text{Enc}(\text{pk}, x)$ and set $c_i = \langle x, s_i \rangle + m_i^{(0)} \bmod 2$ for all $i \in [\delta]$. Output $\text{ct} = (c_0, s_1, \dots, s_\delta, c_1, \dots, c_\delta)$.

Partial Decryption Queries: as in Game₃

Game₅ :

Encryption Queries: as in Game₄

Partial Decryption Queries: For each of the ℓ circuits C and corresponding challenge messages $m_1^{(1)}, \dots, m_k^{(1)}$, compute $\text{ct} \leftarrow \text{Eval}'(\text{pk}, C, \text{ct}_1, \dots, \text{ct}_k)$ and then $d_i \leftarrow \text{PartDec}'(\text{sk}_i, \text{ct})$ for all $i \in [n]$. Output $\mathbf{d} = (d_i)_{i \in [n]}$.

Claim. Assume there is an adversary $\mathcal{A}$ of circuit size s' who wins the selective- ℓ -IND-CPA game against ThFHE' with probability at least p . Then, there exists an $i \in \{0, \dots, 4\}$ such that $\text{cdist}_{s'}(\text{Game}_i, \text{Game}_{i+1}) > (2p - 1)/5 := \tilde{\epsilon}$.

Proof. By assumption, it yields

$$\begin{aligned} p &< \Pr[\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{selective-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t) = 1] \\ &= \Pr[\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{selective-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t) = 1 \mid b = 1] \cdot \Pr[b = 1] \\ &\quad + \Pr[\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{selective-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t) = 1 \mid b = 0] \cdot \Pr[b = 0] \\ &= \Pr[\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{selective-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t) = 1 \mid b = 1] \cdot \frac{1}{2} \\ &\quad + \left(1 - \Pr[\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{selective-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t) = 0 \mid b = 0]\right) \cdot \frac{1}{2}. \end{aligned}$$

Now, we observe that $\Pr[\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{selective-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t) = 1 \mid b = 1]$ corresponds to $\Pr[\mathcal{A} \text{ outputs 1 in Game}_0]$ and $\Pr[\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{selective-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t) =$

$0 \mid b = 0]$ corresponds to $\Pr[\mathcal{A} \text{ outputs } 1 \text{ in } \text{Game}_5]$. By the definition of cdist this implies $\text{cdist}_{s'}(\text{Game}_0, \text{Game}_5) > (p - \frac{1}{2})2 = 2p - 1$. Using the triangle inequality, there exists $i \in \{0, \dots, 4\}$ such that $\text{cdist}_{s'}(\text{Game}_i, \text{Game}_{i+1}) > (2p - 1)/5 = \tilde{\varepsilon}$. ■

Next, we argue for all $i \in \{0, \dots, 4\}$, if $\text{cdist}_{s'}(\text{Game}_i, \text{Game}_{i+1}) > \tilde{\varepsilon}$, it either breaks one-way security of ThFHE or circuit privacy of ThFHE' . Note that the modifications from Game_0 to Game_1 are the same (in reverse order) as from Game_4 to Game_5 . Similarly, the modifications from Game_1 to Game_2 are the same as from Game_3 to Game_4 . Moreover, Game_2 and Game_3 are information-theoretically close to each other, because the challenge messages $m^{(0)}$ or $m^{(1)}$ are hidden by truly random bits. We thus focus on the step from Game_0 to Game_1 and the step from Game_1 to Game_2 in the following. The step from Game_0 to Game_1 is necessary to correctly apply the Goldreich-Levin extractor argument in the next step. By replacing the evaluation algorithm with the direct encryption of the evaluated circuit, we make sure that partial decryptions do not leak any information on the challenge bit $b = 1$.¹⁰

Claim (Game_0 to Game_1). Assuming that $\text{cdist}_{s'}(\text{Game}_0, \text{Game}_1) > \tilde{\varepsilon}$ contradicts the $(s', \tilde{\varepsilon}/\ell)$ -circuit privacy of ThFHE' .

Proof. As Game_0 and Game_1 only differ on how the ℓ partial decryption queries are answered, it yields

$$\tilde{\varepsilon} < \text{cdist}_{s'}(\text{Game}_0, \text{Game}_1) \leq \ell \cdot \text{cdist}_{s'}(\mathbf{d}, \tilde{\mathbf{d}}),$$

where $\mathbf{d}$ is a vector of partial decryptions output in Game_0 and $\tilde{\mathbf{d}}$ a vector of partial decryptions output in Game_1 . Using that applying the randomized function $\text{PartDec}'(\text{sk}_i, \cdot)$ does not increase the computational distance and using the definitions of $\text{PartDec}'$ and Eval' (through PartDec and Eval , respectively), we observe that $\tilde{\varepsilon}/\ell$ is bounded above by

$$\text{cdist}_{s'} \left(((\text{sk}_i)_i, \text{Eval}(\text{pk}, \tilde{C}, \text{ct}_1, \dots, \text{ct}_k)), ((\text{sk}_i)_i, \text{Enc}(\text{pk}, \tilde{C}(x_1^{(1)}, \dots, x_k^{(1)}))) \right),$$

contradicting the $(s', \tilde{\varepsilon}/\ell)$ -circuit privacy of ThFHE (cf. Def. 14). We later link $\tilde{\varepsilon}$ to ε' as in the theorem statement. ■

Claim (Game_1 to Game_2). Assuming that $\text{cdist}_{s'}(\text{Game}_1, \text{Game}_2) > \tilde{\varepsilon}$ contradicts the $(\ell(k+1), \delta)$ -OW-CPA security assumption of ThFHE .

Proof. As Game_1 and Game_2 only differ on how the $k\ell$ encryption queries are answered, it yields

$$\tilde{\varepsilon} < \text{cdist}_{s'}(\text{Game}_1, \text{Game}_2) \leq k\ell \cdot \text{cdist}_{s'}(\text{ct}, \tilde{\text{ct}}),$$

¹⁰ We have overseen this subtlety in an earlier version of this paper [BS23b] and thank the Asiacrypt'23 reviewers for pointing it out to us.

where ct is an encryption output in Game_1 and $\tilde{\text{ct}}$ is an encryption output in Game_2 . We can rewrite ct using the concatenated Goldreich-Levin extractor $\mathcal{E}$ from Definition 2. We define $m := (0, \dots, 0, m^{(1)}) \in \mathcal{M}^\delta \times \mathcal{M}'$, $X := U(\mathcal{M})$ and $Z := (\text{Enc}(\text{pk}, X), (\text{sk}_i)_{i \in S}, \mathbf{E}, \text{PartD})$, the latter being the random variable defined by the randomized encryption algorithm for uniform random messages, the corrupted secret key shares, the circuit evaluations and the partial decryptions given by the partial decryption queries in the security game. Furthermore, we set $Y = X$, such that $\text{cdist}_{s'}((X, Z), (Y, Z)) \leq \varepsilon$ for all $s', \varepsilon > 0$. We observe that $\text{ct} = (Z, \mathcal{E}(X, U(\mathcal{M}^\delta) + m))$ and $\tilde{\text{ct}} = (Z, U(\mathcal{M}^\delta \times \mathcal{M}') + m)$. It holds that

$$\tilde{\varepsilon}/(k\ell) < \text{cdist}_{s'}(\text{ct}, \tilde{\text{ct}}) \leq \text{cdist}_{s'}((Z, \mathcal{E}(X, U(\mathcal{M}^\delta)), (Z, U(\mathcal{M}^\delta \times \mathcal{M}'))).$$

Applying Lemma 2 implies an upper bound on the unpredictability entropy, i.e., $H_{\varepsilon, s}^{\text{unp}}(X|Z) < \lambda$, where $\varepsilon = \frac{\tilde{\varepsilon}}{5k\ell\delta}$, $s = O(s'\gamma^{-3}\varepsilon^4)$ and $\lambda = \delta + \log_2 2\gamma + 3\log_2 1/\varepsilon$.¹¹ To conclude the proof of the claim, we link the unpredictability entropy of X given Z to the OW-CPA security of ThFHE via a reduction. In the following, we explain how the corresponding oracle queries for ThFHE' (which define X and Z) can be answered by having access to the three analogue oracles (denoted OEnc, OChallEnc and OPartDec) from the OW-CPA security game, cf. Definition 17.

Reduction to OW-CPA Game. Let $(\vec{m}_i^{(0)}, \vec{m}_i^{(1)}, C_i)_{i \in [\ell]}$ be the ℓ message vector pairs (each vector of dimension k) and ℓ circuits output by the selective- ℓ -IND-CPA adversary. For simplicity, the reduction always calls the OChallEnc oracle, and never calls the OEnc oracle of the OW-CPA game.

Encryption Queries: Every of the ℓ message vector pairs is composed of k simple message pairs. For every of those $k \cdot \ell$ message pairs $m^{(0)}, m^{(1)} \in \{0, 1\}^\delta = \mathcal{M}'$, query OChallEnc (on no input) and get back an encryption $c_0 = \text{Enc}(\text{pk}, x)$ for an unknown x . For $i \in [\delta]$, sample $s_i \leftarrow U(\mathcal{M})$. Define the circuit $\tilde{C}$ which takes as input x and computes $\langle x, s_i \rangle + m_i^{(1)} \bmod 2$ for every $i \in [\delta]$. Then query OPartDec on c_0 and the circuit $\tilde{C}$. For every $i \in [\delta]$, the partial decryption oracle outputs all partial decryption shares that can be combined to $c_i = \langle x, s_i \rangle + m_i^{(1)}$. Output the ciphertext $\text{ct} = (c_0, s_1, \dots, s_\delta, c_1, \dots, c_\delta)$.

Partial Decryption Queries: For each of the ℓ circuits C and corresponding challenge messages $m_1^{(1)}, \dots, m_k^{(1)}$, define the constant circuit C' which, on any input, simply outputs $C(m_1^{(1)}, \dots, m_k^{(1)})$. Query OPartDec on the circuit C' . On output $\mathbf{d} = (d_i)_{i \in [n]}$ of the oracle OPartDec, output $\mathbf{d}$.

All of the $(k + 1)\ell$ queries to OPartDec done within encryption queries and within partial decryption queries do pass the entropy-check with the entropy

¹¹ Here, we see why the proof in our earlier versions [BS23b; BS23c; BS23e] was flawed: in Lemma 2 the leakage Z is assumed to be independent of the uniform elements in the Goldreich-Levin extractor. We can guarantee this only in a selective-query model.

bound δ (cf. line 10 of Figure 3). Regarding the first case, by Lemma 3, every inner product $\langle x, s_i \rangle \bmod 2$ leaks at most one bit of x . Hence, at most δ bits are leaked in total when querying OPartDec on circuit $\tilde{C}$. Similarly, the circuit C' leaks at most δ bits for every x_i with $i \in [k]$. To conclude the proof, we observe that $H_{\varepsilon,s}^{\text{unp}}(X|Z) < \lambda$ implies that for any adversary $\mathcal{B}$ of circuit size s

$$2^{-\lambda} < \Pr[\mathcal{B}(Z) = X] \leq \text{Adv}_{\text{ThFHE}}^{((k+1)\ell, \delta)\text{-OW-CPA}}(\mathcal{B}).$$

■

Regarding the parameters from the theorem statement, we observe from the above two sub proofs that $\varepsilon' = \tilde{\varepsilon}/\ell = \varepsilon 5k\ell\delta/\ell = \varepsilon 5k\delta$ and $s' = O(s\gamma^3\varepsilon^{-4})$ as stated. $\square$

5 Threshold Fully Homomorphic Encryption From LWE With Polynomial Modulus

We now present our construction of a t -out-of- n ThFHE scheme with OW-CPA security. First, we describe and analyze our main construction based on any LSSS with strong $\{0, 1\}$ -reconstruction. Then, in Section 5.5, we give an alternative construction that combines pseudorandom secret sharing with Shamir sharing to improve efficiency when $\binom{n}{t}$ is small.

By applying the OW-CPA to IND-CPA transformation for ThFHE from Section 4.2, we hence obtain an IND-CPA secure scheme. When we restrict ourselves to standard PKE, our construction gives us a standard ThPKE scheme (cf. Def. 12). We can then also apply the alternative transformation from Section 4.1, which additionally achieves some form of robustness.

5.1 Nearly Linear Decryption of FHE

We use the following abstraction of LWE-based encryption schemes, where decryption is viewed as a linear function of the secret key that outputs a “noisy” version of the correct message. Similar notions were used in [BKS19; Bra+19].

Definition 22 (FHE with (β, ε) -linear decryption). Let $\text{FHE} := (\text{Setup}, \text{Enc}, \text{Dec}, \text{Eval})$ be a fully homomorphic encryption scheme (as in Def 13) with message space $\mathcal{M} \subseteq R_p$ and ciphertext space R_q^r . Suppose that Setup outputs a secret key $\text{sk} \in R_q^r$ which has the form $(1, \mathbf{s})$ for some $\mathbf{s} \in R_q^{r-1}$.

Let $\beta = \beta(\lambda) \in \mathbb{N}, \varepsilon = \varepsilon(\lambda) \in [0, 1]$. We say that FHE has (β, ε) -linear decryption if for any $\lambda, \kappa \in \mathbb{N}$, $(\text{pp}, \text{pk}, \text{sk}) \leftarrow \text{Setup}(1^\lambda, 1^\kappa)$, depth- κ circuit $C: \mathcal{M}^k \rightarrow \mathcal{M}$, messages $m_1, \dots, m_k \in R_p$, ciphertexts $\mathbf{c}_i \leftarrow \text{Enc}(\text{pk}, m_i) \in R_q^r$ and $\text{ct} \leftarrow \text{Eval}(\text{pk}, \mathbf{c}_1, \dots, \mathbf{c}_k)$, it holds that

$$\langle \text{sk}, \text{ct} \rangle = \lfloor q/p \cdot C(m_1, \dots, m_k) \rfloor + e \pmod{q},$$

for some $e \in R_q$ such that $\Pr[\|e\|_\infty \leq \beta] \geq 1 - \varepsilon$ (where the probability is taken over the randomness of Setup, Enc and Eval).

In standard (Module)-LWE based constructions, it's possible to securely set the parameters such that the ratio β/q can be made arbitrarily small, and as long as we have $\beta/q = 1/\text{poly}(\lambda)$, then q is $\text{poly}(\lambda)$.

For security, we require that FHE is IND-CPA secure.¹² This can be instantiated under the Module-LWE assumption to obtain (leveled) FHE using, for instance, the BGV scheme [BGV12] (with superpolynomial q). For $p = 2$, $d = 1$ and $R = \mathbb{Z}$, we also get (leveled) FHE under the standard LWE assumption with a polynomial modulus q [BV14].

5.2 Construction from LSSS with Strong $\{0, 1\}$ -Reconstruction

Our construction works over the ring $R = \mathbb{Z}[X]/f(X)$ for some degree- d irreducible polynomial f , and uses the following main ingredients:

- $\mathcal{D}_{\text{flood}}$: a noise distribution over $\mathbb{Z}_q$ with magnitude bounded by β_{flood} ,
- $\mathcal{D}_{\text{sim}}$: a noise distribution over $\mathbb{Z}_q$, where $\text{RD}_a(\mathcal{D}_{\text{sim}} \| \mathcal{D}_{\text{flood}} + B) \leq \varepsilon_{\text{RD}_a}$, for some $a \in (1, \infty)$, $\varepsilon_{\text{RD}_a} > 1$ and for all B with $|B| \leq \beta_{\text{fhe}}$,
- LSS: a t -out-of- n linear secret sharing scheme $\text{LSS} = (\text{Share}, (\text{Rec}_S)_{S \subseteq [n]})$ with strong $\{0, 1\}$ -reconstruction, associated parameters $L, \tau_{\text{max}}, \tau_{\text{min}}$ and shares in $\mathbb{Z}_q^L$ (cf. Def. 7),
- FHE: a OW-CPA secure $\text{FHE} = (\text{Setup}', \text{Enc}, \text{Eval}, \text{Dec})$ scheme with message space $\mathcal{M} \subseteq R_p$, ciphertext space R_q^r , and $(\beta_{\text{fhe}}, \varepsilon)$ -linear decryption for some $\beta_{\text{fhe}} < q/(2p) - \tau_{\text{min}}\beta_{\text{flood}}$ and some negligible ε .

We now define the scheme $\text{ThFHE} := (\text{Setup}, \text{Enc}, \text{Eval}, \text{PartDec}, \text{Combine})$ by using Enc and Eval from the underlying FHE scheme and setting Setup , PartDec and Combine as specified in Figure 7. We prove its correctness in Appendix C.

Setup ($1^\lambda, 1^\kappa, n, t$)	PartDec (sk_i, ct)
1 : $(\text{pp}, \text{pk}, \text{sk}) \leftarrow \text{Setup}'(1^\lambda, 1^\kappa)$ 2 : $\parallel \text{sk} \in R_q^r, \text{sk}_i \in (R_q^r)^L$ 3 : $(\text{sk}_1, \dots, \text{sk}_n) \leftarrow \text{LSS.Share}(\text{sk})$ 4 : return $(\text{pp}, \text{pk}, \text{sk}_1, \dots, \text{sk}_n)$	1 : $\mathbf{e}_{i,j} \leftarrow \mathcal{D}_{\text{flood}, R_q}$ for $j \in [L]$ 2 : $\parallel \text{sk}_i = (\text{sk}_{i,1}, \dots, \text{sk}_{i,L}) \in (R_q^r)^L$ 3 : $\mathbf{d}_{i,j} \leftarrow \langle \text{ct}, \text{sk}_{i,j} \rangle + \mathbf{e}_{i,j}$ 4 : return $\mathbf{d}_i \leftarrow (\mathbf{d}_{i,1}, \dots, \mathbf{d}_{i,L})$
	Combine ($\{\mathbf{d}_i\}_{i \in S}, \text{ct}$)
	1 : $y \leftarrow \text{Rec}_S((\mathbf{d}_i)_{i \in S})$ 2 : return $\lfloor (p/q) \cdot y \rfloor$

Fig. 7. Setup, partial decrypt and combine algorithms for OW-CPA secure ThFHE. The Enc and Eval algorithms are the same as for FHE.

¹² In our main construction, we assume $\mathcal{M}$ is large and only rely on OW-CPA security of FHE. When extending to smaller $\mathcal{M}$ in Sec. 5.3, we instead need IND-CPA security.

For now, we assume the plaintext space $\mathcal{M} \subseteq R_p$ is superpolynomial in the security parameter, so that FHE is OW-CPA secure. In Section 5.3, we show how to extend this to use FHE with any plaintext space, which allows instantiating from LWE with polynomial modulus.

We write $\mathcal{D}_{\text{flood}, R_q^r}$ (resp. $\mathcal{D}_{\text{sim}, R_q^r}$) to refer to the distribution consisting of rd independent $\mathcal{D}_{\text{flood}}$ (resp. $\mathcal{D}_{\text{sim}}$) random variables, used to sample the coefficients of r elements of R_q .

We show security in the following.

Theorem 4 (Security). *For any adversary $\mathcal{A}$ against the (ℓ, ν) -OW-CPA property of the ThFHE scheme in Fig. 7 with message space $\mathcal{M}$, there exists an adversary $\mathcal{B}$ against the IND-CPA property of FHE, such that*

$$\text{Adv}_{\text{ThFHE}}^{(\ell, \nu)\text{-OW-CPA}}(\mathcal{A}) \leq \left[|\text{ChallM}| \left(\text{Adv}_{\text{FHE}}^{\text{IND-CPA}}(\mathcal{B}) + 2^{-\log_2(|\mathcal{M}|) + \nu} \right) \cdot \varepsilon_{\text{RD}_a}^{\ell d(nL - \tau_{\max})} \right]^{(a-1)/a} + \ell \varepsilon,$$

where L and $\tau_{\max}$ are parameters from the LSS and $|\text{ChallM}|$ is the number of challenge ciphertexts the adversary queried.

Proof. The high-level idea is to modify the (ℓ, ν) -OW-CPA game (Figure 3) such that the t secret shares and the answers to the ℓ partial decryption queries provided to the adversary no longer depend on the underlying secret key sk . This is reflected by the sequence of games from G_0 to G_4 . In the new game G_4 , the adversary still learns the circuit evaluations, stored in the set $\mathbf{E}$, which might leak some information on the challenge messages, stored in ChallM . In a final step, when going to G_5 , we make those circuit evaluations independent of the challenge ciphertexts, by tweaking the oracle OChallEnc to output random ciphertexts (independent of the challenge messages). Here we need to assume the IND-CPA security of the underlying non-threshold FHE scheme. By arguing that the circuit evaluations coming from the partial decryption queries do not leak too much information on the challenge messages, we can bound the advantage of the resulting adversary in the last game G_5 to be negligible.

Game G_0 : This is the real threshold (ℓ, ν) -OW-CPA experiment as in Figure 3. The view of $\mathcal{A}$ is given by

$$\mathcal{V} = (\text{pp}, \text{pk}, \{\text{sk}_i\}_{i \in S}, \text{CT}, \text{ChallCT}, \mathbf{E}, \text{PartD}),$$

where pp are the public parameters, pk is the public key, $\{\text{sk}_i\}_{i \in S}$ are the secret key shares given to the adversary, CT and ChallCT contain the (challenge) ciphertexts the adversary has queried, $\mathbf{E}$ and PartD store the results of up to ℓ adaptive circuit evaluations and partial decryption queries. In each partial decryption query, $\mathcal{A}$ inputs a circuit C and list of indices $(i_1, \dots, i_k)$, and receives $(\mathbf{d}_i)_{i \in [n]}$, where $\mathbf{d}_i$ is the partial decryption of $\text{ct} \leftarrow \text{Eval}(\text{pk}, C, \text{ct}_{i_1}, \dots, \text{ct}_{i_k})$ under sk_i . Once the adversary knows all the partial decryption shares, they can reconstruct the circuit evaluation $C(m_{i_1}, \dots, m_{i_k})$. It yields, $\text{Adv}_{\text{ThFHE}}^{(\ell, \nu)\text{-OW-CPA}}(\mathcal{A}) = \text{Adv}_{\text{ThFHE}}^{G_0}(\mathcal{A})$.

Game G_1 : In this game, we redefine how the partial decryptions are computed. After the adversary chooses the set $S \subset [n]$ of corrupt parties, let

$S_L = \{(i, j)\}_{i \in S, j \in [L]}$ be the corresponding set of share elements. Fix $T \supseteq S_L$ to be a maximal invalid set of share elements. Then, compute the partial decryptions $\mathbf{d}_i$ for a ciphertext $\mathbf{ct}$ as follows:

1. For $(i, j) \in T$, let $\tilde{\mathbf{d}}_{i,j} = \langle \mathbf{ct}, \mathbf{sk}_{i,j} \rangle$;
2. For $(i, j) \in ([n] \times [L]) \setminus T$, let $T_{i,j} \subseteq T \cup \{(i, j)\}$ be a minimal valid set of share elements, and compute $\tilde{\mathbf{d}}_{i,j} = \langle \mathbf{ct}, \mathbf{sk} \rangle - \sum_{(k,l) \in T_{i,j} \setminus \{(i,j)\}} \tilde{\mathbf{d}}_{k,l}$;
3. Sample $\mathbf{e}_i \leftarrow \mathcal{D}_{\text{flood}, R_q^L}$ and compute $\mathbf{d}_i = \tilde{\mathbf{d}}_i + \mathbf{e}_i$, for $i \in [n]$.

Game G_2 : In this game, before outputting the partial decryptions for a ciphertext $\mathbf{ct}$, we first check that $\langle \mathbf{ct}, \mathbf{sk} \rangle = \lfloor q/p \rfloor \cdot C(m_1, \dots, m_k) + e$ for some e with $\|e\|_\infty \leq \beta_{\text{fhe}}$. If not, the game aborts.

Game G_3 : We replace the partial decryptions corresponding to shares outside of T with simulated ones. Firstly, in step (2) above, for $(i, j) \in ([n] \times [L]) \setminus T$, we now compute $\tilde{\mathbf{d}}_{i,j}$ as $\tilde{\mathbf{d}}_{i,j} = \lfloor q/p \rfloor \cdot C(m_1, \dots, m_k) - \sum_{(k,l) \in T_{i,j} \setminus \{(i,j)\}} \tilde{\mathbf{d}}_{k,l}$. Secondly, in step (3), instead of always sampling $\mathbf{e}_{i,j} \leftarrow \mathcal{D}_{\text{flood}, R_q}$, we only sample $\mathbf{e}_{i,j} \leftarrow \mathcal{D}_{\text{flood}, R_q}$ if $(i, j) \in T$, and $\mathbf{e}_{i,j} \leftarrow \mathcal{D}_{\text{sim}, R_q}$ otherwise.

Game G_4 . In the next game, we change how the secret key shares are sampled: pick $(\mathbf{sk}'_1, \dots, \mathbf{sk}'_n) \leftarrow \text{LSS.Share}(0)$ and give to $\mathcal{A}$ the shares $\{\mathbf{sk}'_i\}_{i \in S}$.

Game G_5 . In the last game, we replace the oracle OChallEnc by $\text{OChallEnc}'$, as defined in Figure 8. In the new oracle, two independent m and m' are sampled. Whereas m is added to the challenge message list ChallM , the encryption of m' is added to the challenge ciphertext list ChallCT .

$\text{OChallEnc}'()$	
1 : $\text{idx} = \text{idx} + 1$	
2 : $m, m' \leftarrow \mathcal{M}$	
3 : $\text{ChallM} = \text{ChallM} \cup \{m\}$	
4 : $\mathbf{ct} \leftarrow \text{Enc}(\text{pk}, m')$	
5 : $\text{ChallCT} = \text{ChallCT} \cup \{\mathbf{ct}\}$	
6 : $\text{L}[\text{idx}] := \{(1, m, \mathbf{ct})\}$	
7 : return $\mathbf{ct}$	

Fig. 8. Modified $\text{OChallEnc}'$ oracle.

The theorem then follows from the following lemmata relating the advantages between the different games and showing that the final advantage in the last game is negligibly small.

Lemma 8. *For any PPT adversary $\mathcal{A}$ in Games G_0 and G_1 , it holds that*

$$\text{Adv}_{\text{ThFHE}}^{G_0}(\mathcal{A}) = \text{Adv}_{\text{ThFHE}}^{G_1}(\mathcal{A}).$$

Proof. Note that the view of $\mathcal{A}$ in G_1 is identical to that in G_0 , due to the strong $\{0,1\}$ -reconstruction property of LSS. This is because every share belonging to the maximally invalid set T is computed the same way as in G_0 , using the shares sk_i , while each share outside this set is deterministically fixed to be a sharing of the correct secret $\langle \text{ct}, \text{sk} \rangle$, plus noise sampled from $\mathcal{D}_{\text{flood}}$, as in G_0 . Hence, $\text{Adv}_{\text{ThFHE}}^{G_0}(\mathcal{A}) = \text{Adv}_{\text{ThFHE}}^{G_1}(\mathcal{A})$. $\blacksquare$

Lemma 9. *For any PPT adversary $\mathcal{A}$ in Games G_1 and G_2 , it holds that*

$$\text{Adv}_{\text{ThFHE}}^{G_1}(\mathcal{A}) \leq \text{Adv}_{\text{ThFHE}}^{G_2}(\mathcal{A}) + \ell\varepsilon.$$

Proof. Due to the $(\beta_{\text{fhe}}, \varepsilon)$ -linear decryption property of FHE, and applying a union bound over the ℓ queries, we have that $\text{Adv}_{\text{ThFHE}}^{G_1}(\mathcal{A}) \leq \text{Adv}_{\text{ThFHE}}^{G_2}(\mathcal{A}) + \ell\varepsilon$. $\blacksquare$

Lemma 10. *For any PPT adversary $\mathcal{A}$ in Games G_2 and G_3 , it holds that*

$$\text{Adv}_{\text{ThFHE}}^{G_2}(\mathcal{A}) \leq (\text{Adv}_{\text{ThFHE}}^{G_3}(\mathcal{A}) \cdot \varepsilon_{\text{RD}_a}^{\ell d(nL - \tau_{\max})})^{(a-1)/a},$$

where $\tau_{\max}$ is the size of the smallest maximal invalid share set in LSS.

Proof. We compute the Rényi divergence between the views of the adversary in each game. Each view consists of the adversary's random tape and

$$\mathcal{V} = (\text{pp}, \text{pk}, \{\text{sk}_i\}_{i \in S}, \text{CT}, \text{ChallCT}, \text{E}, \text{PartD}),$$

where CT and ChallCT store the (challenge) ciphertexts and E and PartD the circuit evaluations and partial decryption shares after the ℓ partial decryption queries. For simpler notation, we set $\mathbf{i}^\eta := (i_1^\eta, \dots, i_k^\eta)$ and $\mathbf{m}^\eta := (m_{i_1}^\eta, \dots, m_{i_k}^\eta)$ for the index list and corresponding message vector of the η -th query. Let D_2 and D_3 denote the distributions of $\mathcal{V}$ in games G_2 and G_3 , respectively. Since the partial decryption queries are adaptive, note that the circuit C^η and the index list $\mathbf{i}^\eta$ input during the η -th query depend on the previous queries to OEnc , OChallEnc and OPartDec and the corresponding responses. However, since each $(C^\eta, \mathbf{i}^\eta)$ is a deterministic function of the other values in the view (including the random tape), by the data processing inequality (Lem. 4), $\text{RD}_a(D_2 \| D_3) \leq \text{RD}_a(D'_2 \| D'_3)$, where D'_2, D'_3 are the distributions with the $C^\eta, \mathbf{i}^\eta$ values removed. D'_2 and D'_3 are now defined identically, except in the way the partial decryption components $\mathbf{d}_{i,j}^\eta$ are computed for indices $(i, j) \notin T$. In G_2 , $\mathbf{d}_{i,j}^\eta$ is computed using (amongst other values) $\langle \text{ct}^\eta, \text{sk} \rangle + \mathcal{D}_{\text{flood}, R_q}$, whereas G_3 instead uses $\lfloor q/p \cdot C(\mathbf{m}^\eta) \rfloor + \mathcal{D}_{\text{sim}, R_q}$. Since $\langle \text{ct}^\eta, \text{sk} \rangle = \lfloor q/p \cdot C(\mathbf{m}^\eta) \rfloor + e_\eta$ for some e_η with $\|e_\eta\|_\infty \leq \beta_{\text{fhe}}$, and the view contains $nL - |T|$ pairs $(i, j) \notin T$ where the sampling of $\mathbf{d}_{i,j}^\eta$ changes from G_2 to G_3 , to compute $\text{RD}_a(D'_2 \| D'_3)$, it suffices to compute

$$\text{RD}_a \left(((e_1 + \mathcal{D}_{\text{flood}, R_q})^{nL - |T|}, \dots, (e_\ell + \mathcal{D}_{\text{flood}, R_q})^{nL - |T|}) \| \mathcal{D}_{\text{sim}, R_q}^{\ell(nL - |T|)} \right).$$

Applying Lem. 6 with $N = d\ell(nL - |T|)$, $D_1 = \mathcal{D}_{\text{flood}}$, $D_2 = \mathcal{D}_{\text{sim}}$, we get

$$\text{RD}_a(D'_2 \| D'_3) \leq \varepsilon_{\text{RD}_a}^{d\ell(nL-|T|)}.$$

Applying the probability preservation property of Rényi divergence, we bound the success probability of the adversary as required. $\blacksquare$

Lemma 11. *For any PPT adversary $\mathcal{A}$ in Games G_3 and G_4 , it holds that*

$$\text{Adv}_{\text{ThFHE}}^{G_3}(\mathcal{A}) = \text{Adv}_{\text{ThFHE}}^{G_4}(\mathcal{A}).$$

Proof. Note that the view of $\mathcal{A}$ in G_4 is perfectly indistinguishable from the one in G_3 by the perfect privacy property of LSS. Hence, $\text{Adv}_{\text{ThFHE}}^{G_3}(\mathcal{A}) = \text{Adv}_{\text{ThFHE}}^{G_4}(\mathcal{A})$. $\blacksquare$

Lemma 12. *For any PPT adversary $\mathcal{A}$ in Games G_4 and G_5 , it holds that*

$$\text{Adv}_{\text{ThFHE}}^{G_4}(\mathcal{A}) \leq \text{Adv}_{\text{ThFHE}}^{G_5}(\mathcal{A}) + |\text{ChallM}| \cdot \text{Adv}_{\text{FHE}}^{\text{IND-CPA}}(\mathcal{A}).$$

Proof. The view of $\mathcal{A}$ in G_5 and G_4 are computationally indistinguishable assuming the IND-CPA security of the non-threshold FHE scheme for every query to $\text{OChallEnc}'$. In total, there are $|\text{ChallM}|$ many such queries. Hence, we obtain $\text{Adv}_{\text{ThFHE}}^{G_4}(\mathcal{A}) \leq \text{Adv}_{\text{ThFHE}}^{G_5}(\mathcal{A}) + |\text{ChallM}| \cdot \text{Adv}_{\text{FHE}}^{\text{IND-CPA}}(\mathcal{A})$. $\blacksquare$

Lemma 13. *For any PPT adversary $\mathcal{A}$ in Game G_5 , it yields that*

$$\text{Adv}_{\text{ThFHE}}^{G_5}(\mathcal{A}) \leq q_c \cdot 2^{-\log_2(|\mathcal{M}|)} \cdot 2^\nu,$$

where $\mathcal{M}$ is the message space, ν the bound on the entropy leakage guaranteed in the (ℓ, ν) -OW-CPA game and $q_c := |\text{ChallM}|$ the number of queried ciphertext challenges. If ν is logarithmic, $|\mathcal{M}|$ exponential and q_c polynomial in λ , the advantage is negligible in λ .

Proof. Let $\mathcal{V}$ denote the views of $\mathcal{A}$ in Game G_5 . It is given by

$$\mathcal{V} = (\text{pp}, \text{pk}, \{\text{sk}_i\}_{i \in S}, \text{CT}, \text{ChallCT}, \text{E}, \text{PartD}).$$

Note that in Game G_5 , all challenge messages in ChallM are independent of the challenge ciphertexts in ChallCT . Furthermore, since Game G_4 the secret key shares $\{\text{sk}_i\}_{i \in S}$ of the corrupted parties are shares of zero, and by this independent of the secret key sk as well as independent of the challenge messages in ChallM . Regarding the partial decryption shares stored in PartD , we observe that since Game G_3 , they are simulated using the evaluated plaintext messages stored in E and the decryption shares of corrupted parties. The public parameters pp , public key pk and normal ciphertexts stored in CT , are trivially independent of ChallM . Thus, $\tilde{H}_\infty(m|\mathcal{V}) = \tilde{H}_\infty(m|\text{E})$. Overall, it yields

$$\begin{aligned} \text{Adv}_{\text{ThFHE}}^{G_5}(\mathcal{A}) &\leq \sum_{m \in \text{ChallM}} 2^{-\tilde{H}_\infty(m|\mathcal{V})} = \sum_{m \in \text{ChallM}} 2^{-\tilde{H}_\infty(m|\text{E})} \\ &\leq \sum_{m \in \text{ChallM}} 2^{-\tilde{H}_\infty(m) + \nu}, \end{aligned}$$

where we used that the leakage is guaranteed to be bounded above by ν . Finally, we use that every $m \in \text{ChallM}$ is sampled uniformly at random over $\mathcal{M}$, thus $\tilde{H}_\infty(m) = \log_2(|\mathcal{M}|)$, leading to $\text{Adv}_{\text{ThFHE}}^{G_5}(\mathcal{A}) \leq |\text{ChallM}| \cdot 2^{-\log_2(|\mathcal{M}|)} \cdot 2^\nu$. $\blacksquare$

□

5.3 Supporting a Larger Plaintext Space

The above construction works for a plaintext space $\mathcal{M} \subseteq R_p$. Since we only obtain one-way security, this requires $|R_p|$ to be superpolynomial in λ to give a meaningful security guarantee. If R_p is small, we can easily modify our threshold scheme to still be secure by using several ciphertexts to encrypt larger messages with the underlying FHE scheme. Note that this change is necessary to obtain an instantiation from LWE with polynomial modulus, since there $\mathcal{M} = R_p = \mathbb{Z}_2$.

Concretely, suppose that FHE is IND-CPA secure and has small message space $\mathcal{M}$. Define FHE' with message space $\mathcal{M}^k$, such that $|\mathcal{M}^{-k}|$ is negligible, by encrypting each of the k message components separately under FHE. We then instantiate our threshold scheme using FHE' instead of FHE, where during the partial decrypt and combine steps, we run the algorithms for the previous construction on each component separately. If FHE is IND-CPA secure, then so is FHE', and the proof carries over in the same way, except that the ℓ values in the statement of Theorem 4 will be replaced with $k\ell$, to account for the fact that each of the ℓ decryption queries involves k decryptions of ciphertexts from FHE.

5.4 Bounding the Rényi Divergence

We now analyze parameters and instantiate the distributions $\mathcal{D}_{\text{flood}}$ and $\mathcal{D}_{\text{sim}}$. For now, we simply choose them both to be rounded Gaussian distributions $[D_\sigma]$ with the same standard deviation σ . In Sec. 6.1, we obtain tighter parameters by carefully optimizing the choice of distributions. If FHE has a maximum ciphertext noise bound of β_{fhe} , then using Lem. 5 with our choice of distributions, we get $\varepsilon_{\text{RD}_a} = \text{RD}_a(\mathcal{D}_{\text{flood}} + \beta_{\text{fhe}} \|\mathcal{D}_{\text{sim}}\|) \leq \exp\left(\frac{a\beta_{\text{fhe}}^2}{2\sigma^2}\right)$. If FHE has λ_{FHE} bits of security, then from Thm. 4, the resulting ThFHE scheme is λ_{ThFHE} -bit secure, such that

$$\lambda_{\text{ThFHE}} \geq (\lambda_{\text{FHE}} - \ell d(nL - \tau_{\max}) \log_2 \varepsilon_{\text{RD}_a}) \frac{a-1}{a} \quad (1)$$

Combining the above two equations, we obtain $\lambda_{\text{ThFHE}} \geq \frac{a-1}{a} \lambda_{\text{FHE}} - \ell d(nL - \tau_{\max})(a-1) \frac{\beta_{\text{fhe}}^2}{2\sigma^2} \log_2 e$. Setting for instance $a = \lambda_{\text{ThFHE}}$, and choosing $\sigma, q, \beta_{\text{fhe}}$ such that $\sigma = O(\beta_{\text{fhe}} \sqrt{\ell d(nL - \tau_{\max})(a-1)})$ while decryption is still correct, the loss in security is only a constant factor. Smaller values of a give different tradeoffs between the size of σ and the security loss. Note that in any case, if ℓ and nL are polynomially bounded then both σ and the modulus q can be also.

5.5 Alternative Construction Using Pseudorandom Secret Sharing

We also give a different construction based on pseudorandom secret sharing (PRSS), which improves upon the previous one in some aspects. Instead of having each party perturb their share by an independent, random noise term, we will use PRSS [GI99; CDI05]. This allows them to jointly sample replicated secret sharings of small noise terms, without interaction, after a one-time setup that distributes PRF keys. We also exploit the fact that replicated secret shares can be locally converted to any other LSS, and convert the secret shared noise terms into Shamir sharings before using them for partial decryption. This means that the partial decryptions are Shamir shares, which are much smaller, consisting of only 1 element over R_q each. Furthermore, this leads to improved parameters in the security reduction (by avoiding the $nL - \tau_{\max}$ term in Equation 1), and we can additionally take advantage of the error-correction capability of Shamir to achieve strong robustness (Def. 16) when $t < n/3$. This offers a way of getting robustness for ThFHE instead of only ThPKE with our previous transformations, with the drawback that we require $\binom{n}{t}$ to be not too large, due to using replicated secret sharing.

To sum up, PRSS is a lightweight tool for achieving robustness with a small number of parties. The details and security proof of this construction are in Appendix D.

6 Sample Parameters and Security Estimates

In this section, we discuss how to choose concrete parameters for our OW-CPA secure threshold construction, where we take as a starting point the lattice-based scheme Kyber [Sch+20]. Hence, we are not in the fully homomorphic case, but in the standard PKE case and thus obtain a standard ThPKE scheme. We denote the thresholdized version of Kyber by TKyber.

After deriving sample parameter sets in Section 6.1, we give in Section 6.2 an attack if the adversary has access to sufficiently many partial decryptions. We will see that the bound is close to the one obtained in Section 5, showing that using the Rényi divergence leads to almost optimal results.

We recall the high level description of Kyber in App. E. The relevant parameters for Kyber are the ring degree d , the rank r , the modulus q and the width η of the secret key and encryption randomness distributions. Whereas the specifications of Kyber only consider three parameter sets, called Kyber512, Kyber768 and Kyber1024, we additionally consider three more parameter sets, that we subsequently call Kyber1280, Kyber1536 and Kyber1792. As the name suggest, they are obtained in a similar manner as the previous parameter sets, simply by increasing the rank by +1. All parameter sets are summarized in Table 4 in Appendix E.

6.1 Security From the Reduction

Let λ_{PKE} (resp. λ_{ThPKE}) denote the security level of the starting PKE (resp. the resulting ThPKE) from Theorem 4. Further, we set $\Delta_\lambda := \lambda_{\text{PKE}} - \lambda_{\text{ThPKE}}$,

which describe the security loss in our reduction. Instantiating Equation 1 in the standard PKE setting yields

$$\lambda_{\text{ThPKE}} \geq \frac{a-1}{a} \cdot (\lambda_{\text{PKE}} - \ell d(nL - \tau_{\max}) \log_2 \varepsilon_{\text{RD}_a}), \quad (2)$$

where ℓ is the number of partial decryption queries, d the degree of the ring R , L and $\tau_{\max}$ parameters of the underlying LSSS and $\varepsilon_{\text{RD}_a}$ an upper bound on the Rényi divergence $\text{RD}_a(\mathcal{D}_{\text{sim}} \parallel \mathcal{D}_{\text{flood}} + \beta_{\text{pke}})$ of order a . Here, $\mathcal{D}_{\text{sim}}$ (resp. $\mathcal{D}_{\text{flood}}$) denotes the simulating (resp. flooding) noise distribution and β_{pke} is a bound on the decryption noise that depends on the concrete parameters of Kyber, in particular on the ring degree d , the module rank r and the parameter η , as well as the maximal failure probability ε we want to achieve. For concreteness we set λ_{PKE} as the core-SVP classical hardness, i.e., the resulting BKZ block estimated from the Lattice Estimator [APS15] size multiplied by 0.292.

Table 2 and Table 3 present some sample parameters. We explain in Appendix E in more details how we concretely derived them. The relevant difference between the two is that in the first table, we focus on larger numbers of parties n and samples ℓ while accepting a modulus of up to 39 bits. For simplicity, we assume that both $\mathcal{D}_{\text{flood}}$ and $\mathcal{D}_{\text{sim}}$ follow a Gaussian distribution of width σ . In contrast, in the second table we fine-tuned the flooding and simulation distributions so that we can allow for very small q (only multiplying the original Kyber modulus by small constants up to 10).

Table 2. Sample parameters and security estimates following the reduction from Thm. 4 using a generic approach.

Set	$(\beta_{\text{pke}}, \varepsilon)$	n	t	ℓ	$\lceil \log_2 \sigma \rceil$	$\lceil \log_2 q \rceil$	λ_{PKE}	λ_{ThPKE}	Δ_λ
TKyber1024	$(390, 2^{-60})$	2	1	1	17	23	120	117	3
TKyber1024	$(934, 2^{-300})$	2	1	1	18	24	111	108	3
TKyber1024	$(390, 2^{-60})$	10	9	1	17	25	105	102	3
TKyber1280	$(435, 2^{-60})$	10	5	1	21	29	120	117	3
TKyber1536	$(476, 2^{-60})$	20	10	10	27	36	112	109	3
TKyber1792	$(513, 2^{-60})$	2	1	2^{32}	33	39	123	120	3

Table 3. Sample parameters and security estimates following the reduction from Thm. 4 obtained from a hand-tuned Python program.

Set	q	n	t	ℓ	$\mathcal{D}_{\text{flood}}$	$\mathcal{D}_{\text{sim}}$	λ_{ThPKE}	Δ_λ
TKyber1024	$5 \cdot 3329$	2	1	1	947	1087	100	111
TKyber1024	$10 \cdot 3329$	2	1	2	1994	2034	104	91
TKyber1024	$9 \cdot 3329$	3	2	1	1197	1297	106	92

6.2 Statistical Attack

In the following, we describe an attack against our proposed threshold decryption scheme if the adversary obtains sufficiently many partial decryption queries. Note that the obtained lower bound on the samples for this attack is only slightly higher than the upper bound for security from Section 5. This shows that using the Rényi divergence leads to quasi optimal parameters.

As in the previous section, we focus on Kyber and denote by TKyber the thresholdized scheme as in Section 5. For simplicity, we consider the full-threshold setting for n parties using additive secret sharing. We use as flooding noise distribution a rounded Gaussian $\lfloor \mathcal{D}_{\text{flood}, R_q} \rfloor$ of width σ_{flood} .

Lemma 14. *Let q, d, r, η be the Kyber parameters (as introduced in App. E). Further, let ℓ denote the number of partial decryption queries to TKyber an adversary $\mathcal{A}$ has access to. Further, let $\nu \in \mathbb{N}$. If*

$$\ell d = \Omega((2r+1)d + \nu) \quad \text{and} \quad \ell d = \Omega\left(\frac{\sigma_{\text{flood}}^2}{\eta^2} \log_2(2d(2r+1))\right),$$

then $\mathcal{A}$ can recover the secret key of TKyber with probability $1 - 1/2d(2r+1) - 2^{-\nu}$.

Proof. As we use additive secret sharing, every party receives exactly one secret key share sk_i , where $\text{sk} = \sum_{i=1}^n \text{sk}_i$.

Following the description of Kyber from App. E and the threshold function from Figure 7, a partial decryption of TKyber is of the form $d = (d_i)_{i \in [n]}$, with

$$d_i = v \cdot 1_i - \mathbf{u}^T \mathbf{s}_i + e_i,$$

where 1_i is a share of 1 (e.g. $1_i = 1$ if $i = 1$ and 0 otherwise) and $e_i \leftarrow \mathcal{D}_{\text{flood}, R_q}$.

Without loss of generality, we say that Party 1 is honest and all other parties are controlled by the adversary $\mathcal{A}$. After receiving all n decryption shares, the adversary can sum them up to obtain

$$\sum_{i=1}^n d_i = \mathbf{r}^T \mathbf{e} - \mathbf{e}_1^T \mathbf{s} + e_2 + \lfloor q/2 \rfloor m + \sum_i e_i,$$

where $(\mathbf{r}, \mathbf{e}_1, e_2)$ is the encryption randomness used for this query.

We can re-write $\sum_i d_i = \langle \mathbf{w}, \mathbf{z} \rangle + \lfloor q/2 \rfloor m + \sum_i e_i$, where $\mathbf{w} = (\mathbf{r}, \mathbf{e}_1, e_2)^T \leftarrow \text{CBD}_{\eta}^{(2r+1)d}$ and $\mathbf{z} = (\mathbf{e}, -\mathbf{s}, 1)^T$.

After subtracting $\lfloor q/2 \rfloor m$, the adversary obtains $d' = \langle \mathbf{w}, \mathbf{z} \rangle + \sum_{i=1}^n e_i$. Moreover, the adversary knows the flooding noise of the corrupted parties and can further subtract it from d' , leading to $d'' = \langle \mathbf{w}, \mathbf{z} \rangle + e_1$.

Interestingly, we observe that all elements appearing in the equation of d'' are of small norm, thus no reduction modulo q is necessary. After applying the coefficient embedding, we can interpret d'' as d samples of I-LWE as defined in Section 2.3. Due to the concrete shape of $R_q = \mathbb{Z}_q[X]/(X^d + 1)$ in Kyber, the resulting public matrix $\mathbf{W}$ of the I-LWE instance is now the concatenation

of nega-cyclic matrices over $\mathbb{Z}_q$. Overall, after ℓ partial decryption queries, the adversary has seen an instance of the I-LWE distribution of parameters $R := (2r+1)d$ and $M := \ell d$ with underlying secret $\mathbf{z} \in \mathbb{Z}^R$. Recall that in TKyber, the distribution of $\mathbf{w}$ is given by a centered binomial distribution of parameter η , defining a η -subgaussian distribution with $\sigma_w = \sqrt{\mathbb{E}[\chi_e^2]} \leq \sqrt{\eta^2} = \eta$. The noise follows a rounded Gaussian distribution, is thus σ_{flood} -subgaussian. Thus, Theorem 1 leads to an attacker with success probability $1 - 1/2R - 2^{-\nu}$ if $M = \Omega((2r+1)d + \nu)$ and $M = \Omega\left(\frac{\sigma_{\text{flood}}^2}{\eta^2} \log_2(2d(2r+1))\right)$. Here we use that the least square method performs for $\mathbf{W}$ (with the nega-cyclic structure) as good as for matrices where every entry is independent of all the others. That is the case, as the nega-cyclic structure preserves the required properties to prove Theorem 1. $\square$

In comparison, in Section 5.4 we require $M = \ell d = O\left(\frac{\sigma_{\text{flood}}^2}{\beta_{\text{fhe}}^2}\right)$. Recall that β_{fhe} is the bound on the ciphertext noise, which depends on the decryption failure probability one wants to tolerate. Some concrete parameters for TKyber are given in Table 2. In all cases, $\beta_{\text{fhe}} \geq \eta / \log_2(2d(2r+1))$ and hence our upper bound from Section 5 is below the lower bound from the attack.

Note that [ASY22] showed that the Rényi divergence in their threshold signature leads to optimal bounds by providing an attack for larger bounds. As they use a deterministic signature scheme, their analysis boils down to a straightforward averaging attack. In our case, we argue with the results on Integer LWE, using the least square method, as our encryption scheme is randomized.

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Appendix A Missing Definitions of Section 3

A.1 Compactness and Decryption Correctness

We define the properties of compactness and decryption correctness in the following. Note that compactness is only relevant in the fully homomorphic setting.

Definition 23 (Compactness). *We say that a ThFHE scheme satisfies compactness if there exists a polynomial poly such that for all λ, κ, n, t, C with $C: \mathcal{M}^k \rightarrow \mathcal{M}$ a circuit of depth at most κ and for all $(m_j)_{j \in [k]} \in \mathcal{M}^k$ the following holds. For $(\text{pp}, \text{pk}, \text{sk}_1, \dots, \text{sk}_n) \leftarrow \text{Setup}(1^\lambda, 1^\kappa, n, t)$, $\text{ct}_j \leftarrow \text{Enc}(\text{pk}, m_j)$ for $j \in [k]$ and $\text{ct} \leftarrow \text{Eval}(\text{pk}, C, \text{ct}_1, \dots, \text{ct}_k)$, it yields*

$$|\text{ct}| \leq \text{poly}(\lambda, \kappa, n),$$

where $|\text{ct}|$ denotes the bit size of ct .

Definition 24 (Decryption Correctness). *We say that a ThFHE scheme satisfies decryption correctness if there exists a negligible function $\text{negl}(\lambda)$ such that for all $\lambda, \kappa, n, t, S, C$ with $S \subset [n]$ of size at least $t+1$ and $C: \mathcal{M}^k \rightarrow \mathcal{M}$ of depth at most κ , and for all $(m_j)_{j \in [k]} \in \mathcal{M}^k$ the following holds. For $(\text{pp}, \text{pk}, \text{sk}_1, \dots, \text{sk}_n) \leftarrow \text{Setup}(1^\lambda, 1^\kappa, n, t)$, $\text{ct}_j \leftarrow \text{Enc}(\text{pk}, m_j)$ for $j \in [k]$, $\text{ct} \leftarrow \text{Eval}(\text{pk}, C, \text{ct}_1, \dots, \text{ct}_k)$ and decryption shares $d_i \leftarrow \text{PartDec}(\text{sk}_i, \text{ct})$ for $i \in S$, it holds*

$$\Pr[\text{Combine}(\{d_i\}_{i \in S}, \text{ct}) = C(m_1, \dots, m_k)] = 1 - \text{negl}(\lambda).$$

A.2 More on Game-Based Indistinguishability

In the following, we present a third flavor of game-based ℓ -IND-CPA security for ThFHE, which we used in the first version of this paper [BS23a]. Compared to the other two notions already presented in Section 3.4, this one is much weaker, which is why we call it **weak- ℓ -IND-CPA**.

Weak Indistinguishability. In this version of indistinguishability, the adversary can only query partial decryptions of *freshly* encrypted ciphertexts, cf. Line 5 of OPartDec. Hence, those ciphertexts are completely independent of the provided challenge ciphertext.

Definition 25 (weak- ℓ -IND-CPA for ThFHE). *A ThFHE scheme is said to fulfill weak- ℓ -IND-CPA security for the security parameter λ , the circuit depth bound κ , the threshold parameters n, t and the query bound ℓ , if for all PPT adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3, \mathcal{A}_4)$*

$$\text{Adv}_{\text{ThFHE}}^{\text{weak-}\ell\text{-IND-CPA}}(\mathcal{A}) := \left| \Pr[\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{weak-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t) = 1] - \frac{1}{2} \right| = \text{negl}(\lambda),$$

where $\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{weak-}\ell\text{-IND-CPA}}$ is the experiment in Fig. 9 with $\text{ctr} = 0$ at the beginning.

$\text{Expt}_{\mathcal{A}, \text{ThFHE}}^{\text{weak-}\ell\text{-IND-CPA}}(1^\lambda, 1^\kappa, n, t)$	$\text{OPartDec}(C, m_1, \dots, m_k)$
1 : $(\text{pp}, \text{pk}, \text{sk}_1, \dots, \text{sk}_n) \leftarrow \text{Setup}(1^\lambda, 1^\kappa, n, t)$	1 : $\text{ctr} = \text{ctr} + 1$
2 : $S \leftarrow \mathcal{A}_1(\text{pp}, \text{pk}) : S \subset [n] \wedge S \leq t$	2 : if $\text{ctr} > \ell$ then return $\perp$
3 : $\text{state} \leftarrow \mathcal{A}_2^{\text{OPartDec}}(\text{pp}, \text{pk}, \{\text{sk}_i\}_{i \in S})$	3 : if $(m_j)_j \notin \mathcal{M}^k$ then return $\perp$
4 : $b \leftarrow U(\{0, 1\})$	4 : if $\text{depth}(C) > \kappa$ then return $\perp$
5 : $(m_0, m_1) \leftarrow \mathcal{A}_3(\text{pp}, \text{pk}, \{\text{sk}_i\}_{i \in S})$	5 : $\text{ct}_j \leftarrow \text{Enc}(\text{pk}, m_j), \forall j \in [k]$
6 : $\text{ct}_b \leftarrow \text{Enc}(\text{pk}, m_b)$	6 : $\rho = \text{randomness used for Enc}$
7 : $b' \leftarrow \mathcal{A}_4^{\text{OPartDec}}(\text{pk}, \{\text{sk}_i\}_{i \in S}, \text{ct}_b)$	7 : $\text{ct} \leftarrow \text{Eval}(\text{pk}, C, \text{ct}_1, \dots, \text{ct}_k)$
8 : return $b = b'$	8 : $d_i \leftarrow \text{PartDec}(\text{sk}_i, \text{ct}), i \in [n]$
	9 : return $\rho, (d_i)_{i \in [n]}$

Fig. 9. Experiment weak- ℓ -IND-CPA security of ThFHE schemes.

Appendix B Missing Proofs of Section 4

B.1 Missing Proofs of Section 4.1

Lemma 15 (Decryption Correctness). *The scheme ThPKE' of Section 4.1 satisfies decryption correctness, if ThPKE satisfies decryption correctness and $\delta = \text{poly}(\lambda)$.*

Proof. Fix λ, n, t, S with $S \subset [n]$ of size at least $t + 1$ and let $m \in \mathcal{M}'$. Compute $(\text{pp}, \text{pk}, \text{sk}_1, \dots, \text{sk}_n) \leftarrow \text{Setup}'(1^\lambda, n, t)$ and $\text{ct} \leftarrow \text{Enc}'(\text{pk}, m)$. For $i \in S$ we denote by $(d_{ij})_{j \in [\delta]} = \mathbf{d}_i \leftarrow \text{PartDec}'(\text{sk}_i, \text{ct})$ the decryption shares. The inequality $\text{Combine}'(\{\mathbf{d}_i\}_{i \in S}, \text{ct}) \neq m$ holds if for at least one $j \in [\delta]$ the inequality $\text{Combine}(\{d_{ij}\}_{i \in S}, c_j) \neq x_j$ is true. By the union bound we have

$$\begin{aligned}
\Pr[\text{Combine}'(\{\mathbf{d}_i\}_{i \in S}, \text{ct}) = m] &= 1 - \Pr[\text{Combine}'(\{\mathbf{d}_i\}_{i \in S}, \text{ct}) \neq m] \\
&= 1 - \Pr\left[\bigcup_{j \in [\delta]} \text{Combine}(\{d_{ij}\}_{i \in S}, c_j) \neq x_j\right] \\
&\leq 1 - \delta \cdot \text{negl}(\lambda) = 1 - \text{negl}(\lambda),
\end{aligned}$$

when $\delta = \text{poly}(\lambda)$. $\square$

B.2 Missing Proofs of Section 4.2

Lemma 16 (Compactness). *The scheme ThFHE' of Section 4.2 satisfies compactness if ThFHE satisfies compactness and $\delta, \gamma = \text{poly}(\lambda, \kappa, n)$.*

Proof. It yields $|\text{ct}| = |\text{c}_0| + (\gamma + 1)\delta$. From the compactness of ThFHE follows that $|\text{c}_0| \leq \text{poly}(\lambda, \kappa, n)$ and hence the claim follows. $\square$

Lemma 17 (Decryption Correctness). *The scheme ThFHE' of Section 4.2 satisfies decryption correctness if ThFHE satisfies decryption correctness.*

Proof. Fix $\lambda, \kappa, n, t, S, C'$ with $S \subset [n]$ of size at least $t+1$ and $C': (\mathcal{M}')^k \rightarrow \mathcal{M}'$ of depth at most κ . Further, let $(m_j)_{j \in [k]} \in (\mathcal{M}')^k$. Compute $(\text{pp}, \text{pk}, \text{sk}_1, \dots, \text{sk}_n) \leftarrow \text{Setup}'(1^\lambda, 1^\kappa, n, t)$, $\text{ct}_j \leftarrow \text{Enc}'(\text{pk}, m_j)$ for $j \in [k]$ and $\text{ct} \leftarrow \text{Eval}(\text{pk}, C', \text{ct}_1, \dots, \text{ct}_k)$. Then,

$$\begin{aligned} & \Pr [\text{Combine}'(\{d_i\}_{i \in S}, \text{ct}) = C'(m_1, \dots, m_k)] \\ &= \Pr [\text{Combine}(\{d_i\}_{i \in S}, c_0) = C(x_1, \dots, x_k)] \\ &= 1 - \text{negl}(\lambda), \end{aligned}$$

where C is defined as in Eval' . $\square$

Appendix C Missing Proofs of Section 5

Theorem 5. *The construction in Fig. 7 satisfies decryption correctness.*

Proof. Let $S \subset [n]$ be of size $> t$, and ct be a ciphertext output from Eval on input a set of honestly generated ciphertexts and a circuit C of depth $\leq \kappa$. Let $\mathbf{d}_i \leftarrow \text{PartDec}(\text{sk}_i, \text{ct})$ for $i \in S$, where $(\text{sk}_1, \dots, \text{sk}_n) = \text{Share}(\text{sk})$.

By the strong $\{0, 1\}$ -reconstruction property of LSS and the validity of S , there exists a minimal valid set of share elements $T \subseteq S \times [L]$ such that

$$\text{Rec}_S((\text{sk}_i)_{i \in S}) = \sum_{(i,j) \in T} \text{sk}_{i,j} = \text{sk}.$$

It follows that

$$\begin{aligned} \text{Combine}(\{\mathbf{d}_i\}_{i \in S}, \text{ct}) &= \left\lfloor (p/q) \cdot (\langle \text{ct}, \text{sk} \rangle + \sum_{(i,j) \in T} \mathbf{e}_{i,j}) \right\rfloor \\ &= \left\lfloor (p/q) \cdot \left(\lfloor (q/p)m \rfloor + e_{\text{ct}} + \sum_{(i,j) \in T} \mathbf{e}_{i,j} \right) \right\rfloor \\ &= \left\lfloor (p/q) \cdot \left((q/p)m + e_{\text{rnd}} + e_{\text{ct}} + \sum_{(i,j) \in T} \mathbf{e}_{i,j} \right) \right\rfloor \\ &= m + \left\lfloor (p/q)(e_{\text{rnd}} + e_{\text{ct}} + \sum_{(i,j) \in T} \mathbf{e}_{i,j}) \right\rfloor, \end{aligned}$$

where e_{ct} is the ciphertext error and e_{rnd} is a rounding polynomial with coefficients $\leq 1/2$. Letting $e = e_{\text{rnd}} + \dots$ be the sum of the 3 error terms, by the $(\beta_{\text{fhe}}, \varepsilon)$ -linear decryption property of FHE, except with probability ε , we have $\|e\|_\infty \leq 1/2 + \beta_{\text{fhe}} + |T| \cdot \beta_{\text{flood}}$. Since $\beta_{\text{fhe}} \leq q/(2p) - \tau_{\min}\beta_{\text{flood}} - 1$ and T is a minimal valid set (so $|T| \leq \tau_{\min}$), we have $\|e\|_\infty < q/(2p)$, so the resulting error term rounds to zero, giving the correct message m . $\square$

Appendix D Details on PRSS-based Construction

D.1 Pseudorandom Secret Sharing

Pseudorandom secret sharing (PRSS) [GI99; CDI05] allows parties to non-interactively obtain secret-sharings of pseudorandom values, after a one-time setup phase which distributes PRF keys among the parties. We use a variant of PRSS over the integers, where the parties do not get shares of uniform values, but instead values bounded from a small range (similarly to [BD10]).

Using a PRF $F : \{0, 1\}^\lambda \times \{0, 1\}^* \rightarrow [-B, B] \cap \mathbb{Z}$, the t -out-of- n threshold case works as follows:

- As setup, for each size- t subset $A \subset [n]$, sample $k_A \leftarrow \{0, 1\}^\lambda$. Give k_A to each party P_i , for $i \in [n]$ where $i \notin A$.
- To sample a pseudorandom share on input a nonce v , party P_i computes the shares $s_A = F(k_A, v)$, for each size- t A where $i \notin A$.

The resulting set of shares $\{s_A\}_{|A|=t}$ form a replicated secret sharing of $s = \sum_A s_A$, and we have $|s| \leq B \cdot \binom{n}{t}$. Furthermore, for any collusion of t parties, there is always one share $s_A \in [-B, B]$ that remains unknown.

Converting to Another LSS. A useful property of replicated secret sharing is that replicated shares can be locally converted into any linear secret sharing scheme for the same access structure via a simple linear transformation [CDI05]. We write the procedure of converting a share s_i into a share s'_i for a LSSS as: $s'_i = \text{Convert}_{\text{rep} \rightarrow \text{LSS}}(s_i)$.

D.2 Construction

The construction is shown in Fig. 10. It uses a PRF $F : \{0, 1\}^\lambda \times R_q^r \rightarrow \mathbb{Z} \cap [-\beta_{\text{flood}}, \beta_{\text{flood}}]$, where we require that the outputs of F are indistinguishable from samples from $\mathcal{D}_{\text{flood}}$.¹³

ThFHE.Setup is modified to sample a set of $\binom{n}{t}$ keys k_A and distribute these to the parties in a replicated secret sharing manner. Meanwhile, the secret key of the PKE scheme is shared using standard Shamir sharing. Then, during partial decryption, the parties use the PRF to obtain replicated secret shares of a noise vector. Finally, the parties convert these to Shamir sharings of the same value, exploiting the generality of replicated secret sharing. The **Combine** algorithm is identical to the previous construction, but using Shamir reconstruction.

Correctness. The proof of correctness follows similarly to the proof of Theorem 5. Since the PRF outputs are bounded by β_{flood} , the noise term sampled with pseudorandom secret-sharing is bounded by $\binom{n}{t} \cdot \beta_{\text{flood}}$. After converting this to Shamir shares, the parties obtain a sharing of the same noise term, so decryption succeeds under the same conditions as in Theorem 5, with $\tau_{\min} = \binom{n}{t}$.

¹³ We can use any PRF, and use the resulting pseudorandom bits to sample from $\mathcal{D}_{\text{flood}}$.

ThFHE.Setup($1^\lambda, n, t$)	ThFHE.PartDec($((\mathbf{sk}_i, \mathbf{k}_i), \mathbf{ct})$)
1 : $(\mathbf{pp}, \mathbf{pk}, \mathbf{sk}) \leftarrow \text{PKE.KGen}(1^\lambda)$ 2 : $k_A \leftarrow \{0, 1\}^\lambda$, for $A \subset [n], A = t$ 3 : $\mathbf{k}_i \leftarrow (k_A)_{i \notin A}$ 4 : $(\mathbf{sk}_1, \dots, \mathbf{sk}_n) \leftarrow \text{Shamir.Share}(\mathbf{sk})$ 5 : return $(\mathbf{pp}, \mathbf{pk}, (\mathbf{sk}_1, \mathbf{k}_1), \dots, (\mathbf{sk}_n, \mathbf{k}_n))$	1 : $\parallel \mathbf{k}_i = (\mathbf{k}_A)_{i \notin A}$, for all $ A = t$ 2 : $e_A \leftarrow F(\mathbf{k}_A, \mathbf{ct})$, for $i \notin A$ 3 : $e_i \leftarrow \text{Convert}_{\text{rep} \rightarrow \text{Shamir}}((e_A)_{i \notin A})$ 4 : $\parallel e_i \in R_q$ 5 : return $d_i \leftarrow \langle \mathbf{ct}, \mathbf{sk}_i \rangle + e_i$

Fig. 10. Setup and partial decrypt algorithms for the variant of the OW-CPA threshold PKE/FHE scheme using pseudorandom secret sharing.

Security. We show security in the following theorem. Note that we improve the security loss compared with Theorem 4, since there is no longer an $nL - \tau_{\max}$ term in the exponent of $\varepsilon_{\text{RD}_a}$.

Theorem 6. *For any adversary $\mathcal{A}$ against the (ℓ, ν) -OW-CPA property of the ThFHE scheme in Fig. 10, there exists an adversary $\mathcal{B}$ against the OW-CPA property of PKE, such that*

$$\text{Adv}_{\text{ThFHE}}^{(\ell, \nu)\text{-OW-CPA}}(\mathcal{A}) \leq \left[q_c \left(\text{Adv}_{\text{PKE}}^{\text{IND-CPA}}(\mathcal{B}) + 2^{-\log_2(|\mathcal{M}|) + \nu} \right) \cdot \varepsilon_{\text{RD}_a}^{\ell d} \right]^{(a-1)/a} + \ell \varepsilon$$

Proof. The proof follows a similar structure to that of Theorem 4, so we only highlight the main differences.

Recall that Game G_0 is the construction. In Game G_1 , we changed the way the partial decryptions were computed, for all shares outside of a maximally invalid set of share elements. Since we now only need to simulate partial decryptions of Shamir shares, we instead define a maximally invalid set of *parties*, $T \supset S$, where S is the set of corrupted parties and T has size t . We then simulate the partial decryptions as follows:

1. For $i \in T$, honestly compute $e_A \leftarrow F(\mathbf{k}_A, \mathbf{ct})$, for each size- t set $A \subset [n]$ with $i \notin A$, and let $d_i = \langle \mathbf{ct}, \mathbf{sk}_i \rangle + \text{Convert}_{\text{rep} \rightarrow \text{Shamir}}((e_A)_{i \notin A})$
2. Sample $e_T \leftarrow \mathcal{D}_{\text{flood}, R_q}$
3. Compute $e = \sum_{A, |A|=t} e_A$
4. For $i \notin T$, let $T' = T \cup \{i\}$ and compute

$$d_i = \lambda_{T', i}^{-1} \cdot \left(\langle \mathbf{ct}, \mathbf{sk} \rangle + e - \sum_{j \in T} \lambda_{T', j} d_j \right)$$

where $\lambda_{T', j}$ are the reconstruction coefficients for Shamir secret sharing, defined by the Lagrange basis for polynomial interpolation at points in T' .

Note that the d_i shares for $i \notin T$ are computed such that the partial decryptions form a valid Shamir sharing of $\langle \mathbf{ct}, \mathbf{sk} \rangle + e$. This is exactly as in the

real protocol, except that here the one share e_T that is not part of any shares in the maximally invalid set T is sampled from $\mathcal{D}_{\text{flood}}$ (step 2) instead of with the PRF. Since the PRF key k_T is not given to the adversary, this hybrid is indistinguishable from the real game G_0 , by the security of the PRF.

Game G_2 then makes the same change as in Theorem 4, removing the possibility of decryption failure. This is indistinguishable from the previous game, except with probability $\ell\varepsilon$.

In Game G_3 , in the noise term e , the share e_T sampled in step 2 is sampled with simulated noise using $\mathcal{D}_{\text{sim}, R_q}$. At the same time, we remove the ciphertext noise term in $\langle \text{ct}, \text{sk} \rangle$, so instead of the last step above, we will now compute

$$d_i = \lambda_{T', i}^{-1} \cdot \left(\lfloor (q/p) \cdot m \rfloor + e - \sum_{j \in T} \lambda_{T', j} d_j \right)$$

Notice that the difference between games G_2 and G_3 is that G_2 uses the real ciphertext noise and $e_T \leftarrow \mathcal{D}_{\text{flood}, R_q}$ to simulate the missing partial decryptions, while G_3 instead uses zero ciphertext noise and $e_T \leftarrow \mathcal{D}_{\text{sim}, R_q}$. Let $e_{\text{ct}} = \langle \text{ct}, \text{sk} \rangle - \lfloor (q/p) \cdot m \rfloor$ be the ciphertext noise. Using Lemma 6, we have

$$\text{RD}_a(\mathcal{D}_{\text{flood}, R_q} + e_{\text{ct}} \| \mathcal{D}_{\text{sim}, R_q}) \leq \varepsilon_{\text{RD}_a}^d$$

Similarly to the proof of Theorem 4, for ℓ decryption queries we obtain

$$\text{Adv}_{\text{ThFHE}}^{G_2}(\mathcal{A}) \leq \left(\text{Adv}_{\text{PKE}}^{G_3}(\mathcal{A}) \cdot \varepsilon_{\text{RD}_a}^{\ell d} \right)^{(a-1)/a}$$

and the result follows. $\square$

Achieving Strong Robustness. An advantage of this construction is that if $t < n/3$, we can exploit the error-correction properties of Shamir sharing to guarantee that **Combine** outputs the correct message, even in the presence of t maliciously chosen partial decryptions. This is because a properly generated **PartDec** output is a valid Shamir share, so the parties can always use Reed-Solomon error correction to reconstruct the secret and decrypt, given at least $n/3$ valid shares. This allows the construction to satisfy the strong chosen-plaintext robustness property (Def. 16). While this is also possible to achieve using the OW-CPA to IND-CPA transformation from Section 4 (and even with $t < n/2$), by using Shamir we avoid the $\binom{n}{t}$ cost of finding the correct subset of partial decryptions, significantly improving the efficiency of the **Combine** algorithm. Furthermore, the Shamir approach is compatible with FHE and not just PKE.

Appendix E More Details on Parameters of Section 6

We recall the high level design of Kyber with messages of the form $m \in R_2 \cong \{0, 1\}^d$, where d denotes the degree of the ring R . The scheme uses the centered binomial distribution with parameter $\eta \in \mathbb{N}$, denoted by CBD_η . We say that

a ring element is sampled from CBD_η if all its d coefficients are independently sampled from CBD_η . This generalizes to vectors in R^r , where r is the underlying module rank. Let $\text{Kyber} = (\text{Setup}, \text{Enc}, \text{Dec})$ be as follows:

Setup(1^λ): Sample short vectors $\mathbf{s}, \mathbf{e} \in R_q^r$ from CBD_η and a uniform matrix $\mathbf{A} \in R_q^{r \times r}$. Set $\text{sk} = (\mathbf{s}, \mathbf{e})$ and $\text{pk} = (\mathbf{A}, \mathbf{t})$, where $\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}$.
Enc(pk, m): Sample a short vector $\mathbf{r} \in R_q^r$ from CBD_η and $\mathbf{e}_1 \in R_q^r$ and $e_2 \in R_q$ from CBD_η . Set $\mathbf{u} = \mathbf{A}^T \mathbf{r} + \mathbf{e}_1$ and $v = \mathbf{r}^T \mathbf{t} + e_2 + \lfloor q/2 \rfloor \cdot m$. Output $\text{ct} = (\mathbf{u}, v)$.
Dec(sk, ct): Compute $c' = v - \mathbf{u}^T \mathbf{s} = \mathbf{r}^T \mathbf{e} - \mathbf{e}_1^T \mathbf{s} + e_2 + \lfloor q/2 \rfloor \cdot m$. Output $\lfloor c' \cdot 2/q \rfloor$.

For simplicity, we omit the additional rounding usually applied to ciphertexts to further reduce their size.

Table 4. Parameter sets for Kyber.

Set	d	r	q	η
Kyber768	256	3	3329	2
Kyber1024	256	4	3329	2
Kyber1280	256	5	3329	2
Kyber1536	256	6	3329	2
Kyber1792	256	7	3329	2

Generic Parameters for Large Numbers of Parties. We first describe a simplified way of deriving parameters, where we assume that $\mathcal{D}_{\text{sim}}$ and $\mathcal{D}_{\text{flood}}$ both are uncut rounded Gaussian distributions of the same width σ .

Using Lemma 5 with our choice of distributions, Equation 2 simplifies to

$$\lambda_{\text{ThPKE}} \geq \frac{a-1}{a} \cdot \left(\lambda_{\text{PKE}} - \ell d(nL - \tau_{\max}) \frac{a\beta_{\text{pke}}^2}{2\sigma^2} \log_2 e \right). \quad (3)$$

When setting $\sigma = \beta_{\text{pke}} \sqrt{\ell d(nL - \tau_{\max})(a-1) \log_2 e}$, the above simplifies to

$$\lambda_{\text{ThPKE}} \geq \frac{a-1}{a} \cdot \lambda_{\text{PKE}} - 1, \quad (4)$$

which promises a rather small security loss at the expense of a larger modulus. Note that we have to set $q > 4(\beta_{\text{pke}} + \tau_{\min}\beta_{\text{flood}})$ in order to guarantee correctness (Thm. 5). Let's for concreteness set $\beta_{\text{flood}} = 10\sigma$ and $a = 100$. Recall that Kyber is a PKE with $(\beta_{\text{pke}}, \varepsilon)$ -linear decryption, where β_{pke} depends on the maximal failure probability ε we tolerate. If we take as a concrete example Kyber1024, it offers $(390, 2^{-60})$ as well as $(934, 2^{-300})$ -linear decryption.

When considering full threshold, we use additive secret sharing and when considering non-full threshold, we assume naive secret sharing, defining the parameters $L, \tau_{\max}, \tau_{\min}$ as in Table 1. After having set σ and q , one can use the

Lattice Estimator [APS15] to derive λ_{PKE} . For simplicity we set λ_{PKE} as the core-SVP classical hardness, i.e., the resulting BKZ block size multiplied by 0.292. The resulting λ_{ThPKE} and Δ_λ then come from Equation 4. We give some sample parameters for TKyber1024 in Table 2. Note that we mean by TKyber1024 that we take all the original Kyber1024 parameters, but modify the modulus q .

Hand-Tuned Parameters for Small Number of Parties. We now describe how we can obtain tighter concrete parameters (in particular a small modulus q) by allowing for different flooding and simulating Gaussian distributions and optimizing their concrete width. Throughout this section, we set $\mathcal{D}_{\text{sim}}$ (resp. $\mathcal{D}_{\text{flood}}$) as the rounded Gaussian distribution of width σ_{sim} (resp. σ_{flood}), where we additionally apply a tail cut after $2 \cdot \sigma_{\text{sim}}$ (resp. $2 \cdot \sigma_{\text{flood}}$).

By extending the Python program for computing security estimates of Kyber¹⁴, we design a Python program that proceeds in the following three steps:

Step 1: Finding $\mathcal{D}_{\text{flood}}$. The high level idea is to find the largest σ_{flood} we can use in our ThPKE such that we still guarantee correctness (Theorem 5). This is how we optimally make use of our modulus q . For simplicity, we set $p = 2$ and hence correctness is fulfilled as long as the infinity norm of the final noise is at most $q/4$. This procedure depends on the Kyber parameters (that define the noise from the decryption algorithm) as well as the maximal decryption failure probability we want to aim for. We fix this probability to be 2^{-60} . At the end, the procedure outputs σ_{flood} and the bound B .

Step 2: Finding $\mathcal{D}_{\text{sim}}$. Once we have computed $\mathcal{D}_{\text{flood}}$, we can find $\mathcal{D}_{\text{sim}}$ such that the Rényi divergence $\text{RD}_2(\mathcal{D}_{\text{flood}} + B \| \mathcal{D}_{\text{sim}})$ is smallest. We start by setting $\mathcal{D}_{\text{sim}} = \mathcal{D}_{\text{flood}}$ and compute the Rényi divergence of order 2. We now (slightly) increase $\mathcal{D}_{\text{sim}}$ step by step and expect the Rényi divergence to decrease up to some optimal sweet spot. Once we observe that the Rényi divergence increases again, we stop increasing $\mathcal{D}_{\text{sim}}$ and take this as the optimal choice. Note that for fixed $\mathcal{D}_{\text{flood}}$, B and $\mathcal{D}_{\text{sim}}$, it yields $\text{RD}_2(\mathcal{D}_{\text{flood}} + B \| \mathcal{D}_{\text{sim}}) \leq \text{RD}_a(\mathcal{D}_{\text{flood}} + B \| \mathcal{D}_{\text{sim}})$ for all $a > 1$. Hence, it is reasonable to compute the sweet spot for the order 2.

Step 3: Finding $\varepsilon_{\text{RD}_a}$. As we now have $\mathcal{D}_{\text{flood}}$, $\mathcal{D}_{\text{sim}}$ and B , we can find the optimal order of the Rényi divergence. Note that, even though $\varepsilon_{\text{RD}_a}$ doesn't decrease for increasing a , the factor $(a-1)/a$ in Equation 2 suggests that the optimal a might not necessarily be $a = 2$. For concreteness, we search the minimum among the orders $a \in [2, \dots, 11]$. We then output the optimal choice of a together with the resulting Rényi divergence $\varepsilon_{\text{RD}_a}$. Finally, we have everything together to compute the upper bound on λ_{ThPKE} .

Table 3 summarizes our findings. We use as base security λ_{PKE} the core-SVP classical hardness of the underlying LWE instance, which can be easily computed using any LWE estimator. For convenience, we used the leaky LWE

¹⁴ <https://github.com/pq-crystals/security-estimates>

estimator [Dac+20]. We give some estimates for the final security λ_{ThPKE} for different choices of small numbers of parties n , threshold t and number of queries ℓ . For all computations, we apply a (rather aggressive) Gaussian tail cut after 2 times the Gaussian width and assume a failure probability bound of 2^{-60} .

Here, we consider variants of the Kyber1024 parameter set, where we multiply the modulus q by some scaling factor. This scaling factor is intended to give an idea of the order of magnitude of the modulus we need. We remark that multiples of 3329 might not necessarily be the optimal choice when taking implementation characteristics into account.

Comparing The Rényi Divergences. We would like to highlight that the two strategies assume different flooding and simulating noise distributions $\mathcal{D}_{\text{flood}}$ and $\mathcal{D}_{\text{sim}}$. Whereas in the first we assume the *same* and (quasi) *uncut* rounded Gaussian distributions, we computed the parameters in the second case with a *different* and *tail cut* rounded Gaussian distributions. When fixing a maximal decryption failure probability, one can choose the modulus q much smaller in the latter case. However, the sharper we cut off the rounded Gaussian distribution, the more the Rényi divergences from Lemma 5 and one computed by our Python program diverge from each other.

E.1 Proof of Lemma 14

Proof. As we use additive secret sharing, every party receives exactly one secret key share sk_i , where $\text{sk} = \sum_{i=1}^n \text{sk}_i$.

Following the description of Kyber from App. E and the threshold function from Figure 7, a partial decryption of TKyber is of the form $d = (d_i)_{i \in [n]}$, with

$$d_i = v \cdot 1_i - \mathbf{u}^T \mathbf{s}_i + e_i,$$

where 1_i is a share of 1 (e.g. $1_i = 1$ if $i = 1$ and 0 otherwise) and $e_i \leftarrow \mathcal{D}_{\text{flood}, R_q}$.

Without loss of generality, we say that Party 1 is honest and all other parties are controlled by the adversary $\mathcal{A}$. After receiving all n decryption shares, the adversary can sum them up to obtain

$$\sum_{i=1}^n d_i = \mathbf{r}^T \mathbf{e} - \mathbf{e}_1^T \mathbf{s} + e_2 + \lfloor q/2 \rfloor m + \sum_i e_i,$$

where $(\mathbf{r}, \mathbf{e}_1, e_2)$ is the encryption randomness used for this query.

We can re-write $\sum_i d_i = \langle \mathbf{w}, \mathbf{z} \rangle + \lfloor q/2 \rfloor m + \sum_i e_i$, where $\mathbf{w} = (\mathbf{r}, \mathbf{e}_1, e_2)^T \leftarrow \text{CBD}_{\eta}^{(2r+1)d}$ and $\mathbf{z} = (\mathbf{e}, -\mathbf{s}, 1)^T$.

After subtracting $\lfloor q/2 \rfloor m$, the adversary obtains $d' = \langle \mathbf{w}, \mathbf{z} \rangle + \sum_{i=1}^n e_i$. Moreover, the adversary knows the flooding noise of the corrupted parties and can further subtract it from d' , leading to $d'' = \langle \mathbf{w}, \mathbf{z} \rangle + e_1$.

Interestingly, we observe that all elements appearing in the equation of d'' are of small norm, thus no reduction modulo q is necessary. After applying the

coefficient embedding, we can interpret d'' as d samples of I-LWE as defined in Section 2.3. Due to the concrete shape of $R_q = \mathbb{Z}_q[X]/(X^d + 1)$ in Kyber, the resulting public matrix $\mathbf{W}$ of the I-LWE instance is now the concatenation of nega-cyclic matrices over $\mathbb{Z}_q$. Overall, after ℓ partial decryption queries, the adversary has seen an instance of the I-LWE distribution of parameters $R := (2r + 1)d$ and $M := \ell d$ with underlying secret $\mathbf{z} \in \mathbb{Z}^R$. Recall that in TKyber, the distribution of $\mathbf{w}$ is given by a centered binomial distribution of parameter η , defining a η -subgaussian distribution with $\sigma_w = \sqrt{\mathbb{E}[\chi_e^2]} \leq \sqrt{\eta^2} = \eta$. The noise follows a rounded Gaussian distribution, is thus σ_{flood} -subgaussian. Thus, Theorem 1 leads to an attacker with success probability $1 - 1/2R - 2^{-\nu}$ if $M = \Omega((2r + 1)d + \nu)$ and $M = \Omega\left(\frac{\sigma_{\text{flood}}^2}{\eta^2} \log_2(2d(2r + 1))\right)$. Here we use that the least square method performs for $\mathbf{W}$ (with the nega-cyclic structure) as good as for matrices where every entry is independent of all the others. That is the case, as the nega-cyclic structure preserves the required properties to prove Theorem 1. $\square$